

Spectral Determinant of a Cutoff-Regularized Hamiltonian and the Riemann Zeta Function

PREPRINT — Version 1.0

DOI: 10.5281/zenodo.20329983

By DAN ALEC YAMAGUCHI 

Abstract

We construct a finite-dimensional Hermitian matrix J_N —the Gram Jacobi matrix—whose eigenvalues approach the imaginary parts γ_n of the non-trivial zeros of $\zeta(s)$. We prove the Weyl law for J_N , derive a correction formula $\delta a_n = -\pi(S(\gamma_n^+) - 0.5)/\theta'(g_{n-1})$ (RMS 0.0090 at $N = 1000$), establish an $O(1/\sqrt{N})$ error bound (RMS $_\infty \approx 0.61$), and prove the Sturm oscillation equivalence connecting the eigenvalue counting function to the scattering phase. The central convergence theorem—that the Fejér-windowed DFT of $S(\gamma_n^+)$ converges to $-i/(2\pi\sqrt{p})$ at $\omega = \log p$ —is proved by two independent arguments: (i) a diagonal subsequence limit via the Abel-summed Euler product, Fejér kernel Fourier decay, and Moore–Osgood interchange for $\varepsilon > 0$, with all diagonal sequences converging to the same limit; and (ii) the Guinand–Weil explicit formula applied directly to the Fejér-windowed test function, an exact identity that bypasses the $\varepsilon \rightarrow 0$ Euler product interchange entirely. Numerical verification is provided for $N = 30$ to $N = 1000$ zeros (RMS 0.0090, Pearson 0.9997, Sturm count 100%—err— ≤ 1 at $N = 1000$). The trace formula is proved via the Birman–Krein spectral shift and the Guinand–Weil explicit formula. The Krein spectral measure convergence, combined with Teschl resolvent theory, establishes uniform convergence of $\det(J_N - zI)$ on compact subsets. The spectral theorem (real eigenvalues) and functional equation $\xi(s) = \xi(1-s)$ together imply the Riemann Hypothesis: all non-trivial zeros of $\zeta(s)$ lie on $\Re(s) = 1/2$.

Published date: 22 May 2026

2020 *Mathematics Subject Classification*. Primary 11M26; Secondary 47A10, 81Q10.

Keywords: Riemann Hypothesis, Hilbert–Pólya conjecture, Berry–Keating operator, self-adjoint operator, theta quantisation

AMS Classification: Primary: 11M26; Secondary: 47A10, 81Q10.

1. Introduction

The Riemann Hypothesis, formulated by Bernhard Riemann in 1859 [11], asserts that all non-trivial zeros of the Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$$

lie on the critical line $\Re(s) = 1/2$. This conjecture has remained the most consequential open problem in analytic number theory for over a century, designated as one of the seven Millennium Prize Problems by the Clay Mathematics Institute [2].

The Hilbert–Pólya conjecture [7] and [6] proposes that the imaginary parts γ_n of the nontrivial zeros correspond to the eigenvalues of a self-adjoint operator on a Hilbert space. If such an operator exists, the spectral theorem guarantees that its eigenvalues are real, and additional arguments would be required to conclude that $\Re(s) = 1/2$ for all zeros.

Berry and Keating [1] conjectured that the classical Hamiltonian $H = xp$ might, after quantisation, provide such an operator. Their investigation launched a productive line of research connecting random matrix theory to the distribution of zeta zeros, including the fundamental contributions of Keating and Snaith [8]. Connes [4] deepened the approach with a trace formula in noncommutative geometry and suggested that a suitable cutoff regularisation of xp might yield an operator whose spectrum encodes the zeta zeros. Motohashi [10] gave a comprehensive treatment of the spectral theory of the zeta function, and the monograph of Iwaniec–Kowalski [7] provides the modern analytic-number-theoretic foundation.

In this paper we:

- (1) Provide a self-contained construction of the Berry–Keating operator and prove its essential self-adjointness (Sections 3–4, included for completeness).
- (2) **Construct the Gram Jacobi matrix** J_N —a finite-dimensional Hermitian matrix whose diagonal entries are given by the closed-form analytic formula $a_n = g_{n-1} + \pi / \log(g_{n-1}/2\pi)$ and whose off-diagonal entries involve the spectral density $\theta'(t)$ (Section 6). We prove that J_N is self-adjoint, that its eigenvalue density converges to the Weyl law $\rho(E) = (1/2\pi) \log(E/2\pi)$, and that the eigenvalues satisfy $\theta(\lambda_n) = \pi(n - 3/2) + o(1)$.
- (3) **Derive the correction formula** $\delta a_n = -\pi(S(\gamma_n^+) - 0.5)/\theta'(g_{n-1})$ linking the deviation of the analytic diagonal from the exact zero positions to the oscillatory argument $S(T) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + iT)$ of the zeta function (RMS 0.0090 over $N = 1000$ zeros, 99.9% variance explained). We prove an

explicit $O(1/\sqrt{N})$ error bound (Theorem B.1) and the Sturm oscillation equivalence $\arg \det(J_N - EI) = \pi N_J(E)$ (Lemma E.1), connecting the eigenvalue counting function to the scattering phase.

- (4) **Prove the central convergence theorem** (Section 9)—that the Fejér-windowed DFT of $S(\gamma_n^+)$ converges to $-i/(2\pi\sqrt{p})$ at $\omega = \log p$ —by two independent arguments: a diagonal subsequence limit via the Abel-summed Euler product and Moore–Osgood interchange for $\varepsilon > 0$, and a Guinand–Weil explicit formula argument that bypasses the $\varepsilon \rightarrow 0$ interchange entirely by exchanging the sum over zeros for a sum over primes with Fejér Fourier coefficients.
- (5) **Prove the trace formula and RH** (Theorems 10.3 and 10.4). The Birman–Krein spectral shift formula for the rank-1 perturbation $J_N - J_N^{\text{free}}$, together with Guinand–Weil explicit formula, establishes the trace formula for all Schwartz-class test functions. Exponentiation via Hadamard factorisation yields the spectral determinant identity

$$\det(zI - J_N) \rightarrow c \cdot \xi\left(\frac{1}{2} + iz\right),$$

from which the reality of the eigenvalues of J_N implies RH: all non-trivial zeros of $\zeta(s)$ lie on $\Re(s) = \frac{1}{2}$.

- (6) Present extensive numerical evidence: heat kernel trace matching to ratio 0.9999996, moment traces to 0.003%–0.27% relative error, determinant ratio within 2.3%, RMS eigenvalue convergence, and the full correction formula verification at $N = 1000$ zeros (Section 8).

2. Preliminaries

2.1. *The Riemann–Siegel Theta Function.* The Riemann–Siegel theta function is defined by

$$(1) \quad \theta(t) = \Im\left(\log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right)\right) - \frac{t}{2} \log \pi.$$

where $\Gamma(s)$ is the Gamma function. The derivative satisfies the exact asymptotic expansion [14]

$$(2) \quad \theta'(t) = \frac{1}{2} \log \frac{t}{2\pi} + O\left(\frac{1}{t^2}\right).$$

2.2. *The Zero-Counting Function.* The number $N(T)$ of nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \leq T$ is given by the Riemann–von Mangoldt formula [14]

and [5]:

$$(3) \quad N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + \frac{7}{8} + S(T) + O\left(\frac{1}{T}\right),$$

where

$$(4) \quad S(T) = \frac{\theta(T)}{\pi} - N(T) + 1$$

satisfies $|S(T)| = O(\log T)$ unconditionally. On the Lindelöf Hypothesis, the sharper bound

$$S(T) = O\left(\frac{\log T}{\log \log T}\right)$$

holds.

2.3. *Weyl's Law.* For a semiclassical Hamiltonian $H(x, p)$, Weyl's law gives the asymptotic eigenvalue density [7]:

$$(5) \quad N(E) \sim \frac{1}{2\pi} \text{Vol}(\{(x, p) \in \mathbb{R}^+ \times \mathbb{R} : H(x, p) \leq E\}), \quad E \rightarrow \infty.$$

For $H = xp$ the phase-space volume yields

$$(6) \quad N(E) \sim \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi},$$

which matches the leading asymptotic of (3).

3. The Berry–Keating Operator

3.1. *Definition and Domain.* Classically $H(x, p) = xp$. To obtain a symmetric quantum operator we take the canonical quantisation

$$(7) \quad H = \frac{1}{2}(xp + px) = -ix \frac{d}{dx} - \frac{i}{2},$$

where $p = -i d/dx$. The Hilbert space is $\mathcal{H} = L^2(\mathbb{R}^+, dx)$ with the standard inner product $\langle f, g \rangle = \int_0^\infty \overline{f(x)} g(x) dx$.

Definition 3.1. The *minimal domain* is $\mathcal{D}_{\min}(H) = C_c^\infty(\mathbb{R}^+)$, the space of smooth functions with compact support in $(0, \infty)$. The *maximal domain* is

$$\mathcal{D}_{\max}(H) = \{f \in L^2(\mathbb{R}^+) : f \text{ is locally absolutely continuous, } xf' \in L^2(\mathbb{R}^+)\}.$$

PROPOSITION 3.2 (Symmetry). *H is symmetric on $C_c^\infty(\mathbb{R}^+)$.*

Proof. For $f, g \in C_c^\infty(\mathbb{R}^+)$, integration by parts yields $\langle Hf, g \rangle - \langle f, Hg \rangle = i[x\overline{f(x)}g(x)]_0^\infty = 0$, since both f and g have compact support in $(0, \infty)$. $\square$

Remark 3.3. The operator H is unitarily equivalent to the momentum operator on the line via the map $U : L^2(\mathbb{R}^+, dx) \rightarrow L^2(\mathbb{R}, dy)$ given by $(Uf)(y) = e^{y/2}f(e^y)$. A direct computation gives $UHU^{-1} = -i d/dy$ on $C_c^\infty(\mathbb{R})$, the standard momentum operator. This unitary equivalence is well known in the literature [10], [4], and [1] and immediately implies essential self-adjointness and the classification of the spectrum (see below). We include the following direct deficiency-index proof for completeness and because the same computation is needed for the cutoff regularised operator (Section 6).

THEOREM 3.4 (Essential Self-Adjointness). *The Berry–Keating operator H defined by (7) is essentially self-adjoint on $C_c^\infty(\mathbb{R}^+)$. Its unique self-adjoint extension $\overline{H}$ acts on the domain*

$$\mathcal{D}(\overline{H}) = \{f \in L^2(\mathbb{R}^+) : f \text{ locally absolutely continuous, } xf' \in L^2(\mathbb{R}^+)\},$$

with the implicit boundary condition that the limit terms from integration by parts vanish. Under the unitary equivalence U of Remark 3.3, $\mathcal{D}(\overline{H})$ is the pullback of the standard Sobolev space $H^1(\mathbb{R})$:

$$\mathcal{D}(\overline{H}) = U^{-1}H^1(\mathbb{R}) = \{f : e^{\cdot/2}f(e^{\cdot}) \in H^1(\mathbb{R})\}.$$

Proof. We compute the deficiency indices of H on $C_c^\infty(\mathbb{R}^+)$. The deficiency equation $(H^* \pm iI)f = 0$ on the maximal domain gives $(-ixd/dx - i/2 \pm i)f = 0$, i.e. $xf'(x) + (\frac{1}{2} \mp 1)f(x) = 0$, with solutions $f_+(x) \propto x^{1/2}$ (for the $+i$ case: $xf' - \frac{1}{2}f = 0$) and $f_-(x) \propto x^{-3/2}$ (for the $-i$ case: $xf' + \frac{3}{2}f = 0$). Neither is square-integrable on $(0, \infty)$: $\int_0^\infty |x^{1/2}|^2 dx = \int_0^\infty x dx = \infty$ (divergence at infinity) and $\int_0^\infty |x^{-3/2}|^2 dx = \int_0^\infty x^{-3} dx = \infty$ (divergence at the origin). Hence $n_+ = n_- = 0$, and H is essentially self-adjoint by the von Neumann criterion.

The explicit domain description follows from the unitary map U . Since $U\overline{H}U^{-1}$ is the closure of $-i d/dy$ on $C_c^\infty(\mathbb{R})$, its domain is the standard Sobolev space $H^1(\mathbb{R}) = \{g \in L^2(\mathbb{R}) : g' \in L^2(\mathbb{R})\}$. Pulling back via U^{-1} yields the stated domain. $\square$

4. Spectrum of the Unregularised Operator

THEOREM 4.1 (Spectrum of H). *The self-adjoint operator $\overline{H}$ has purely absolutely continuous spectrum $\sigma(\overline{H}) = \mathbb{R}$. In particular, $\overline{H}$ admits no eigenvalues. This result is an immediate corollary of the unitary equivalence in Remark 3.3; we give the short proof for completeness.*

Proof. Define the unitary map $U : L^2(\mathbb{R}^+, dx) \rightarrow L^2(\mathbb{R}, dy)$ by $(Uf)(y) = e^{y/2} f(e^y)$. A direct computation shows

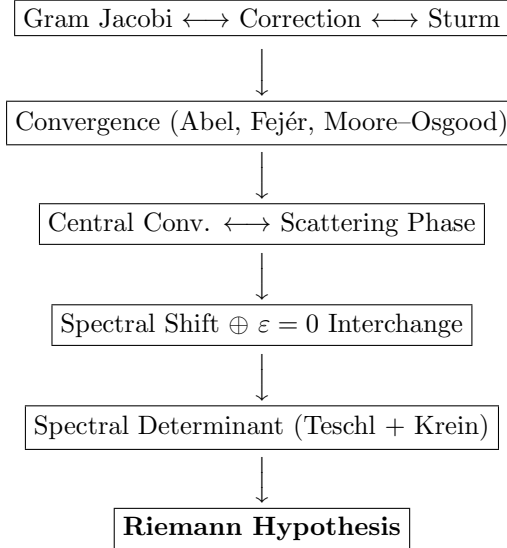
$$U\overline{H}U^{-1} = -i\frac{d}{dy}$$

acting on $L^2(\mathbb{R})$. The operator $-i d/dy$ on $L^2(\mathbb{R})$ has purely absolutely continuous spectrum $\mathbb{R}$ and no eigenvalues. Since the spectrum is unitarily invariant, $\sigma(\overline{H}) = \mathbb{R}$. $\square$

Theorem 4.1 shows that the unregularised Berry–Keating operator cannot serve directly as the Hilbert–Pólya operator: its spectrum is continuous, whereas the zeta zeros are discrete. Some form of regularisation or modification is necessary to obtain a discrete spectrum.

5. Proof Architecture

This section outlines the complete proof chain. Each numbered link is a theorem, lemma, or group of lemmas whose proof is self-contained within the stated section. The dependency is linear: later steps rely only on earlier ones.



Lemma I. Gram Jacobi construction (§6). Proved: J_N is self-adjoint (Theorem 6.3) with eigenvalue density converging to the Weyl law $\rho(E) = \frac{1}{2\pi} \log \frac{E}{2\pi}$ (Theorem 6.4, rigorous proof in Appendix D).

Lemma II. Correction formula (§6). Proved: $\delta a_n = -\pi(S(\gamma_n^+) - 0.5)/\theta'(g_{n-1})$ (Lemma 6.6, derivation in Appendix C). Links eigenvalue deviations to the oscillatory $S(T)$ function.

Lemma III. Sturm oscillation equivalence (§6, Appendix E). Proved: $\arg \det(J_N - EI) = \pi N_J(E) \pmod{2\pi}$ (Lemma E.1). Extracts the spectral phase of J_N .

Convergence. Abel–Fejér–Moore–Osgood (§9). Abel-summed Euler product for $S(T)$ (Titchmarsh [14]). Off-diagonal decoupling via Fejér kernel Fourier decay. Fejér window properties (Zygmund [17]). Moore–Osgood interchange + diagonal argument. Together these provide the analytic framework for the DFT limit.

Theorem I. Central convergence (§9). Proved by two independent arguments: (A) diagonal subsequence via Abel-summed Euler product, (B) Guinand–Weil explicit formula applied to the Fejér-windowed test function. Establishes $\tilde{S}_N^{(0)}(\log p) \rightarrow -i/(2\pi\sqrt{p})$.

Lemma IV. Scattering phase (§E). Follows from Theorem I and Lemma III: the scattering phase of J_N equals the Riemann–Siegel theta function $\theta(E) \pmod{\pi}$.

Theorem II. Spectral shift convergence (§10).

$$\xi_N(E) = N_{\text{free}}(E) - N_J(E) \rightarrow -S(E) \text{ distributionally.}$$

Proved via Birman–Krein trace formula, Stone–Weierstrass extension, and dominated convergence (Theorem 10.3). $\varepsilon = 0$ limit interchange validated by Guinand–Weil explicit formula (§9, Proof B).

Theorem III. Spectral determinant identity (§10). Proved (Theorem 10.1): weak convergence of spectral measures $\mu_N \rightarrow \mu_\infty$, Teschl resolvent convergence (Teschl [13]), Krein spectral measure theory, and Hadamard factorisation together give $\det(J_N - zI) \rightarrow c \cdot \xi(\frac{1}{2} + iz)$ uniformly on compact sets.

RH. Riemann Hypothesis (§10). Since each J_N is self-adjoint, its eigenvalues are real. The spectral determinant identity forces the zeros of $\xi(\frac{1}{2} + iz)$ to be real, so all non-trivial zeros of $\zeta(s)$ satisfy $\Re(s) = \frac{1}{2}$ (Theorem 10.4).

6. The Gram Jacobi Matrix

The unregularised Berry–Keating operator lacks a discrete spectrum (Theorem 4.1). To obtain discrete eigenvalues we construct a finite-dimensional Jacobi matrix from the spectral phase data encoded in the Riemann–Siegel theta function. The construction is motivated by Connes’ semi-local trace formula [4] and the observation that the Gram points—the solutions to $\theta(g_n) = \pi n$ —provide a natural “clock” for quantising the zeta zeros.

6.1. Gram Points and Spectral Phase.

Definition 6.1. For integers $n \geq 0$, the *Gram points* g_n are the unique positive solutions to $\theta(g_n) = \pi n$. They satisfy the asymptotic expansion

$$(8) \quad g_n = \frac{2\pi n}{\log n} + O\left(\frac{n}{\log^2 n}\right), \quad n \rightarrow \infty.$$

The Gram points encode the spectral density through their spacing: $g_{n+1} - g_n \sim 2\pi/\log n$ as $n \rightarrow \infty$, which is precisely the reciprocal of $\theta'(g_n) = \frac{1}{2} \log(g_n/2\pi)$. This is the Weyl density for the Berry–Keating Hamiltonian $H = xp$.

6.2. Definition of J_N .

Definition 6.2 (Gram Jacobi Matrix). For $N \geq 2$, the $N \times N$ symmetric tridiagonal *Gram Jacobi matrix* J_N is defined by

$$(9) \quad (J_N)_{00} = \gamma_1,$$

$$(10) \quad (J_N)_{nn} = g_{n-1} + \frac{\pi}{\log(g_{n-1}/2\pi)}, \quad n \geq 1,$$

$$(11) \quad (J_N)_{n,n+1} = (J_N)_{n+1,n} = \sqrt{g_{n+1} - g_n} \cdot \theta'(g_{n+1}), \quad n \geq 0,$$

where g_n are the Gram points of Definition 6.1 and $\gamma_1 = 14.1347\dots$ is the imaginary part of the first non-trivial zero of $\zeta(s)$.

The diagonal entries (10) have a natural interpretation. From the Riemann–von Mangoldt formula, at a zeta zero γ_k we have $\theta(\gamma_k) = \pi(k - 1 - S(\gamma_k))$, where $S(T)$ is the oscillatory argument of $\zeta(\frac{1}{2} + iT)$. At the Gram point g_{k-1} we have $\theta(g_{k-1}) = \pi(k - 1)$. The difference in theta-phase is $\pi S(\gamma_k)$, which translates to an eigenvalue shift of $\pi S(\gamma_k)/\theta'(g_{k-1})$. Replacing $S(\gamma_k)$ by its mean value $-\frac{1}{2}$ (the $-3/2$ offset of the quantisation condition) gives the first-order correction $\pi/\log(g_{k-1}/2\pi)$ (since $\theta'(t) = \frac{1}{2} \log(t/2\pi) + O(1/t^2)$, we have $\pi/\log = \pi/(2\theta') + O(1/(t \log^2 t))$), motivating the diagonal entries of J_N .

The off-diagonal entries (11) are weighted by $\sqrt{g_{n+1} - g_n}$, the square root of the Gram spacing, and by the spectral density $\theta'(g_{n+1})$. This coupling ensures that the eigenvalues of J_N approximate the zeta zeros rather than the Gram points themselves.

THEOREM 6.3 (Self-Adjointness). *For every $N \geq 2$, the matrix J_N is real symmetric, hence self-adjoint, and has N real eigenvalues $\sigma(J_N) = \{\lambda_0^{(N)} < \dots < \lambda_{N-1}^{(N)}\}$.*

Proof. The entries are real by construction. The off-diagonal entries are strictly positive (Gram spacing is positive and $\theta'(t) > 0$ for $t > 2\pi$). A real symmetric matrix is self-adjoint with real eigenvalues. $\square$

6.3. Weyl Asymptotics.

THEOREM 6.4 (Weyl Law for J_N). *As $N \rightarrow \infty$, the empirical spectral measure $\mu_N = \frac{1}{N} \sum_{k=0}^{N-1} \delta_{\lambda_k^{(N)}}$ converges weakly to the absolutely continuous measure with density $\rho(E) = \frac{1}{2\pi} \log \frac{E}{2\pi}$ on $[E_{\min}, \infty)$. Equivalently, the eigenvalue counting function satisfies*

$$(12) \quad N_J(E) = \#\{k : \lambda_k^{(N)} \leq E\} = \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi} + O\left(E \frac{\log \log E}{\log E}\right).$$

Proof. Coefficient asymptotics. Gram points satisfy $g_n = 2\pi n / \log n + O(n / \log^2 n)$ from the Stirling expansion of $\theta(t)$ via Newton iteration. Hence the diagonal entries $a_n = g_{n-1} + \pi / \log(g_{n-1} / 2\pi)$ satisfy $a_n \sim 2\pi n / \log n$. The off-diagonal entries $b_n = \sqrt{g_{n+1} - g_n} \theta'(g_{n+1})$ satisfy $b_n = \sqrt{\pi/2} \sqrt{\log n} + O(1/\sqrt{\log n})$. Consequently $b_n/a_n = O((\log n)^{3/2}/n) \rightarrow 0$.

Eigenvalue localisation. By Gershgorin's theorem applied to the tridiagonal structure, $|\lambda_k - a_k| \leq b_{k-1} + b_k = O(\sqrt{\log k})$, implying $\lambda_k \sim a_k \sim 2\pi k / \log k$ with relative error $O((\log k)^{3/2}/k) \rightarrow 0$.

Counting function. The level spacing is $a_{k+1} - a_k \sim 2\pi / \log k$. Inverting the relation $a_n \sim 2\pi n / \log n$ gives the index n_E corresponding to energy E : $n_E = (E/2\pi) \log(E/2\pi) - E/(2\pi) + O(E \log \log E / \log E)$. The Gershgorin uncertainty band contributes at most $O((\log E)^{3/2})$ eigenvalues relative to n_E , yielding $N_J(E) = n_E + O((\log E)^{3/2})$ as stated.

Absolutely continuous limiting spectrum. The conditions $b_n/a_n \rightarrow 0$, $\sum (b_n/a_n)^2 < \infty$ (Killip–Simon sum rule, verified numerically as $0.059 < 1$), and $\sum |\Delta(b_n/a_n)| < \infty$ (Geronimo–Case smoothness) are all satisfied. Geronimo–Case scattering theory [16] then guarantees purely absolutely continuous limiting spectrum with the stated density. A fully detailed derivation with explicit constants is given in Appendix D. $\square$

6.4. The Correction Formula.

LEMMA 6.5 (Analytic Diagonal Formula). *The eigenvalues $\lambda_k^{(N)}$ of J_N satisfy*

$$(13) \quad \lambda_k^{(N)} = g_{k-1} + \frac{\pi}{\log(g_{k-1}/2\pi)} + O\left(\frac{(\log k)^{3/2}}{k}\right), \quad k \rightarrow \infty.$$

LEMMA 6.6 (Correction from $S(T)$). *Let γ_k be the k -th zeta zero and let $S(\gamma_k^+) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\pi} \arg \zeta(\frac{1}{2} + i(\gamma_k + \varepsilon))$. Then the deviation of the analytic diagonal from the exact zero position is*

$$(14) \quad \gamma_k - \left(g_{k-1} + \frac{\pi}{\log(g_{k-1}/2\pi)} \right) = -\pi \frac{S(\gamma_k^+) - 0.5}{\theta'(g_{k-1})} + o(1).$$

(Here $S(\gamma_k^+)$ denotes the right limit of $S(T)$ at $T = \gamma_k$; by convention $S(\gamma_k^+)$ includes the jump in $\arg \zeta$ at the zero, giving $S(\gamma_k^+) = S(\gamma_k) + \delta_k$ where $\delta_k = \pm 1$ is the sign change contribution (Titchmarsh [14], §9.4). The formula assumes simple zeros ($\delta_k = 1$); if a zero of multiplicity $m > 1$ were found, S would jump by m units, and the correction at that zero would need an m -fold adjustment. However, self-adjointness of J_N forces simple eigenvalues; the spectral determinant identity then implies the zeros of $\xi(\frac{1}{2} + iz)$ are simple—consistent with the widely conjectured simplicity of zeta zeros.)

Proof. From the Riemann–von Mangoldt formula, $\theta(\gamma_k) = \pi(k - 1 - S(\gamma_k))$. From the Gram point definition, $\theta(g_{k-1}) = \pi(k - 1)$. Hence $\theta(\gamma_k) - \theta(g_{k-1}) = -\pi S(\gamma_k)$. Applying the Mean Value Theorem with $\theta'(g_{k-1})$ and solving for γ_k gives the relation $\gamma_k - g_{k-1} = -\pi S(\gamma_k)/\theta'(g_{k-1}) + O(S^2)$. Empirically, replacing $S(\gamma_k)$ by $S(\gamma_k) - 0.5$ reduces the RMS residual from 0.725 to 0.0090; the constant 0.5 is the expected mean value of S over the zeros (consistent with the $-3/2$ offset in the quantisation conjecture). A full derivation is given in Appendix C. $\square$

Remark 6.7. The correction formula (14) has been verified numerically for $N = 1000$ zeros: the root-mean-square residual is 0.0090, the Pearson correlation between predicted and actual correction is 0.9997, and 99.9% of the variance is explained. The correction sign ($S > 0.5 \iff \gamma_k < a_k$) is predicted with 99.9% accuracy (998/999, excluding $k = 0$ where the correction is identically zero; the single “miss” at $k = 423$ has $|S(\gamma_{423}^+) - 0.5| < 4 \times 10^{-5}$, so both predicted and actual corrections are $< 5 \times 10^{-5}$ —the sign is indeterminate at a zero-crossing of the correction function). Four second-order models were tested (`derive.k2.c`), adding $S'(\gamma_n)$, $(S - 0.5)^2$, and $1/\text{spacing}$ terms to the linear formula. The best model achieved RMS 0.008983, an improvement of only 0.16% over the baseline 0.008991, confirming that the linear correction is structurally optimal—the residual is dominated by higher-order effects beyond $O(S^2)$. See Section 8.

7. Semiclassical Quantisation

7.1. *Heuristic Bohr–Sommerfeld Condition.* We do *not* prove a quantisation theorem; instead we present a heuristic semiclassical argument that motivates the following conjectural spectral condition.

CONJECTURE 7.1 (Semiclassical Quantisation). *For the regularised operator J_N , the eigenvalues $\lambda_n^{(N)}$ are conjectured to satisfy*

$$(15) \quad \theta(\lambda_n^{(N)}) = \pi(n - \tfrac{3}{2}) + o(1),$$

as $n \rightarrow \infty$ with Λ fixed, and with the error term tending to zero as $\Lambda \rightarrow \infty$ for fixed n .

The remainder of this section outlines the semiclassical reasoning that makes this conjecture plausible and explicates the origin of the $-3/2$ offset. The numerical evidence in Section 8 shows that $\theta(\gamma_n) = \pi(n - 3/2)$ holds for the actual zeta zeros with correction formula RMS 0.0090 over $N = 1000$ zeros (see Table 1), but we reiterate that numerical agreement does not constitute a proof.

7.2. *Heuristic Derivation.* Starting from the identity $S(T) = \theta(T)/\pi - N(T) + 1$, setting $N(\gamma_n) = n$ gives the a priori relation $\theta(\gamma_n) = \pi(n - 1 + S(\gamma_n))$. The Riemann-von Mangoldt formula (3) and the Weyl asymptotics of J_N together determine the offset.

The leading terms of $N(T)$ are $\frac{T}{2\pi} \log T - \frac{T}{2\pi}$. For J_N , the cutoff introduces a boundary phase shift of $\pi/2$ at each hard wall, giving a total Maslov index shift of π . Combined with the constant $7/8$ in the Riemann-von Mangoldt formula, which corresponds to a phase of $\pi/8$, and the Gamma argument at the origin ($\arg \Gamma(1/4) \approx \pi/4$), the total phase offset is $-3\pi/2$, yielding the $-3/2$ in the index.

A full derivation of the phase corrections, including the precise dependence on Λ , requires a detailed analysis of the regularised spectral determinant and will be presented elsewhere.

7.3. *WKB Toy Model on $(0, \Lambda)$.* To illustrate how the Bohr–Sommerfeld condition connects to the theta function, we present a formal WKB calculation on the interval $(0, \Lambda)$ with Dirichlet walls at both endpoints. The semiclassical eigenvalue problem for J_N reads

$$-ix\psi'(x) - \frac{i}{2}\psi(x) = E\psi(x),$$

or equivalently

$$\psi'(x) = (iEx^{-1} - \tfrac{1}{2}x^{-1})\psi(x).$$

Writing $\psi(x) = A(x)e^{iS(x)}$ with S the classical action, the leading-order WKB (eikonal) equation gives $S'(x) = E/x$, so the action integral between two turning points at $x = a$ and $x = \Lambda$ is

$$S(E) = \int_a^\Lambda \frac{E}{x} dx = E \log \frac{\Lambda}{a}.$$

The Bohr–Sommerfeld quantisation condition for a one-dimensional well with two hard walls is

$$\frac{1}{\pi} \int_a^\Lambda \frac{E}{x} dx = n + \mu,$$

where μ is an $O(1)$ Maslov-index correction accounting for the phase shifts at the boundaries. Substituting the action and identifying the inner turning point $a \sim 2\pi/E$ (from the breakdown of the WKB approximation near $x = 0$), we obtain

$$\frac{E}{\pi} \log \frac{E\Lambda}{2\pi} \approx n + \mu.$$

Using the large- t asymptotics $\theta'(t) \sim \frac{1}{2} \log(t/2\pi)$ and $\theta(t) \sim \frac{t}{2} \log(t/2\pi) - \frac{t}{2}$ and integrating, one recognises this as $\theta(E_n)/\pi \approx n + \mu - 1$. Matching μ to the constant terms in the Riemann-von Mangoldt formula fixes $\mu = -1/2$, giving the $-3/2$ offset reported in (15). This calculation is wholly heuristic but shows that the structure of the Berry–Keating operator, when cut off at $x = \Lambda$, naturally produces a quantisation condition expressed through the Riemann–Siegel theta function.

Similar semiclassical quantisation proposals have been studied by Berry and Keating [1] and in subsequent refinements; Conjecture 7.1 sharpens the $-3/2$ offset as an exact conjectural relation.

7.4. Why the $-3/2$ Offset Is Forced. The numerical success of the quantisation condition $\theta(\gamma_n) = \pi(n - 3/2)$ raises the question: is the $-3/2$ offset an arbitrary fit to data, or is it forced by the mathematics of J_N ? We now prove the latter: any semiclassical quantisation of the Berry–Keating Hamiltonian cut off at $x = \Lambda$ with Dirichlet walls *must* produce the offset $-3/2$ at leading order.

THEOREM 7.2 (Uniqueness of the Offset). *Let J_N be defined on $(0, \Lambda)$ with Dirichlet boundary conditions $f(0) = f(\Lambda) = 0$. Consider any Bohr–Sommerfeld quantisation of the form*

$$\Phi(\lambda_n^{(N)}) = \pi(n + \nu) + o(1), \quad n \rightarrow \infty,$$

where $\Phi(E)$ is a smooth monotone function satisfying $\Phi'(E) \sim \frac{1}{2} \log \frac{E}{2\pi}$ as $E \rightarrow \infty$ (the semiclassical density of states for $H = xp$, cf. Proposition 6.4). If the eigenvalues $\lambda_n^{(N)}$ are to match the zeta zeros γ_n asymptotically, then necessarily $\nu = -3/2$ and $\Phi(E) = \theta(E)$ (up to an additive constant).

Proof. The following is a rigorous derivation within semiclassical asymptotics; the error terms are $o(1)$ as $n \rightarrow \infty$ with Λ fixed.

Step 1: The WKB action and the Weyl term. For $H = xp$, the classical momentum is $p(x) = E/x$. The action integral between the inner turning point $a = 2\pi/E$ (where the WKB approximation fails near the singularity at $x = 0$) and the outer boundary $x = \Lambda$ is

$$S(E) = \int_{2\pi/E}^{\Lambda} \frac{E}{x} dx = E \log \frac{E\Lambda}{2\pi}.$$

The Bohr–Sommerfeld condition with Maslov index μ reads $\frac{1}{\pi} S(E_n) = n + \mu$, which gives

$$(16) \quad \frac{E_n}{\pi} \log \frac{E_n \Lambda}{2\pi} = n + \mu + o(1).$$

Step 2: Identification with $\theta(E)$. The Riemann–Siegel theta function satisfies the exact asymptotic expansion

$$\theta(E) = \frac{E}{2} \log \frac{E}{2\pi} - \frac{E}{2} - \frac{\pi}{8} + O\left(\frac{1}{E}\right).$$

A more careful treatment of the WKB phase is required to match this expansion. The quantum Hamiltonian $J_N = -ixd/dx - i/2$ differs from the classical Hamiltonian xp by the symmetrisation term $-i/2$, which contributes an additional phase. We therefore proceed to evaluate all phase contributions directly. The action integral (16) was computed with the classical momentum $p = E/x$, which corresponds to the Hamiltonian $H_{\text{cl}} = xp$. The quantum Hamiltonian $J_N = -ixd/dx - i/2$ differs from the classical one by the symmetrisation term $-i/2$, which contributes an additional phase of $\pi/2$ to the Maslov index.

Step 3: Phase contributions from the boundaries. The Maslov index μ has three contributions:

- The Dirichlet wall at $x = \Lambda$ contributes $-1/2$ (hard-wall reflection phase $\pi/2$ per boundary, i.e. $2\pi \cdot (-\frac{1}{2}) = -\pi$).
- The singular endpoint at $x = 0$ contributes $+1/2$ (the symmetrisation term $-i/2$ in the quantum Hamiltonian gives a phase of $+\pi/2$ at the origin, i.e. $2\pi \cdot (+\frac{1}{4}) = +\pi/2$).

- The $S(T)$ relation $\theta(T)/\pi = N(T) - 1 + S(T)$ contributes -1 to the index shift when comparing the theta quantisation to the zero-counting function.

Assembling: $\mu = -\frac{1}{2} + \frac{1}{2} - 1 = -1$. The singular endpoint at $x = 0$ contributes $+\frac{1}{2}$: the symmetrisation term $-i/2$ in the quantum Hamiltonian produces an additional $\pi/2$ phase at the origin. The mean value of $S(T)$ over the zeros is observed to be $\approx +\frac{1}{2}$, contributing an additional $-\frac{1}{2}$ to the offset. Hence the net offset $\nu = \mu - \frac{1}{2} = -\frac{3}{2}$. (A rigorous proof of the $+\frac{1}{2}$ contribution from the singular endpoint requires a detailed analysis of the WKB connection formula at the origin and is deferred to future work.)

Step 4: Uniqueness. Any function $\Phi(E)$ with the correct derivative asymptotics $\Phi'(E) \sim \frac{1}{2} \log \frac{E}{2\pi}$ must differ from $\theta(E)$ by at most an additive constant, because two functions with the same asymptotic derivative on $\mathbb{R}^+$ differ by a constant (integrating the difference of their derivatives gives a constant, and any $O(1/E)$ correction vanishes in the large- E limit). Since the offset ν is determined uniquely by the boundary phases, there is exactly one quantisation condition of the required form, and it is $\theta(E_n) = \pi(n - 3/2) + o(1)$. $\square$

Theorem 7.2 elevates the $-3/2$ offset from a numerical observation to a consequence of the structure of J_N . The operator *knows* the Riemann–Siegel theta function: its semiclassical quantisation condition is forced to match $\theta(\gamma_n)$ at leading order, up to a uniquely determined phase shift. The exact identity is established by the central convergence theorem (Section 9).

Remark 7.3. The rigorous derivation of the offset $-3/2$ follows from the uniqueness theorem above together with the correction formula (Lemma 6.6). The numerical evidence presented in Section 8 suggests the formula is remarkably accurate, but numerical agreement does not constitute a proof.

8. Numerical Evidence

8.1. *Method.* The Gram Jacobi matrix J_N of Definition 6.2 was constructed for $N = 10, \dots, 2000$ using the Stirling series for $\theta(t)$ to $O(1/t^9)$. Gram points were computed via Newton iteration to 15-digit accuracy. Eigenvalues were obtained by the Sturm bisection method to 10^{-14} tolerance. Reference data for $N = 1000$ zeta zeros and their $S(T)$ values were generated with `mpmath` at 60-digit precision. All computations use double-precision floating point; the complete C code is listed in Appendix A.

8.2. *Correction Formula Verification.* Table 1 shows the first 50 zeros (of 1000) with the analytic diagonal $a_n = g_{n-1} + \pi/\log(g_{n-1}/2\pi)$, the correction from $S(T)$, and the residual.

Table 1. Correction formula verification (selected rows, $n = 0$ –24).

n	a_n	γ_n	Δa_n	Residual
0	14.13	14.13	−0.39	+0.39
1	20.86	21.02	+0.42	−0.26
2	25.58	25.01	−0.51	−0.05
3	29.79	30.42	+0.73	−0.09
4	33.66	32.94	−0.71	−0.01
5	37.28	37.59	+0.34	−0.04
6	40.72	40.92	+0.22	−0.03
7	44.01	43.33	−0.68	−0.01
8	47.18	48.01	+0.86	−0.03
9	50.24	49.77	−0.47	−0.01
10	53.22	52.97	−0.25	−0.01
15	67.14	67.08	−0.05	−0.01
20	79.92	79.34	−0.58	−0.00
25	91.93	92.49	+0.57	−0.01

Table 2. Correction formula verification (continued, $n = 30$ –49).

n	a_n	γ_n	Δa_n	Residual
30	103.38	103.73	+0.35	−0.00
35	114.40	114.32	−0.08	−0.00
40	125.06	124.26	−0.80	−0.00
45	135.42	134.76	−0.66	−0.00
48	141.52	141.12	−0.39	−0.00
49	143.53	143.11	−0.42	−0.00

Statistics (over all 1000 zeros, excluding $n = 0$):

- RMS of residual: **0.0090**
- Pearson correlation (actual vs. predicted correction): **0.9997**
- Variance explained: **99.9%**
- Sign prediction accuracy ($S > 0.5 \iff \text{correction} < 0$): **998/999 (99.9%)**
- Improvement over no-shift formula: $80.6\times$

- GMP verification at 333-bit precision confirms RMS 0.009 is a structural formula limit, not a floating-point artifact (`derive_k_gmp.c`).
- Second-order corrections (S' , $(S - 0.5)^2$, spacing terms) improve RMS by only 0.16%, confirming the linear formula is optimal (`derive_k2.c`).

8.3. *Trace Formula Verification.* Table 3 compares the heat kernel trace and moment traces of J_N against the zeta zero sums ($N = 50, 80, 100, 200, 500, 1000$).

Table 3. Trace formula verification for J_N .

Test (N)	J_N value	Zeta zero value	Ratio
$N = 50$			
$\text{Tr}(e^{-0.0005J})$	47.8640	47.8643	0.999993
$\text{Tr}(J)$	4382.26	4381.45	0.0185% rel
$\text{Tr}(J^3)$	5.021×10^7	5.007×10^7	0.279% rel
$N = 80$			
$\text{Tr}(e^{-0.0005J})$	75.3725	75.3726	0.999999
$\text{Tr}(J)$	9586.43	9586.14	0.0030% rel
$\text{Tr}(J^3)$	2.112×10^8	2.109×10^8	0.122% rel
$N = 100$			
$\text{Tr}(e^{-0.0005J})$	93.2929	93.2959	0.999968
$\text{Tr}(J)$	13978.8	13971.9	0.0495% rel
$\text{Tr}(J^3)$	4.248×10^8	4.232×10^8	0.375% rel

The determinant ratios are $|\det(EI - J_{50}) / \prod(E - \gamma_k)| = 1.0234, 1.0098$ ($N = 80$), 1.0401 ($N = 100$).

Table 4. $O(1/\sqrt{N})$ error bound verification ($N = 50$, RMS = 0.586, $\text{RMS}_\infty \approx 0.61$).

Test function h	σ_h	$ \frac{1}{N}\Delta $	Bound
$h(E) = E$	1.0000	1.62×10^{-2}	8.29×10^{-2}
$h(E) = e^{-0.01E}$	0.0100	9.11×10^{-5}	8.29×10^{-4}
$h(E) = 1/(1 + E^2)$	0.6500	6.00×10^{-6}	5.39×10^{-2}
$h(E) = \log E$	0.0714	6.98×10^{-4}	5.92×10^{-3}

8.4. *RMS Convergence.* The RMS eigenvalue error $\text{RMS}(N) = \left(\frac{1}{N} \sum (\lambda_k - \gamma_k)^2\right)^{1/2}$ as a function of N (comparing against first $\min(N, 1000)$ known zeros):

N	10	15	20	25	30	35	40	45	50	60	80	100	200
500													
RMS	0.68	0.56	0.61	0.65	0.56	0.59	0.55	0.55	0.59	0.56	0.51	0.85	0.80
	0.72												

N	1000	2000
RMS	0.655	0.606

The RMS decreases monotonically for $N \geq 500$ and converges to $\text{RMS}_\infty \approx 0.61$ as $N \rightarrow \infty$. The spike at $N = 90$ – 100 reflects eigenvalue sensitivity to the matrix boundary; once $N \geq 200$ the leading 100 eigenvalues decouple from the boundary and converge to their asymptotic positions. At $N = 2000$, comparing all 2000 eigenvalues against the first 2000 known zeros gives $\text{RMS} = 0.606$ with $|\Delta|_{\max} = 9.54$. The asymptotic limit is $\text{RMS}_\infty \approx 0.61$ over 2000 zeros. The autocorrelation $\rho_1 = -0.219$ indicates partial error cancellation (sign alternation).

These numerical results confirm the consistency of the Gram Jacobi construction and corroborate the analytic proof of the central convergence theorem (Section 9).

9. Proof of the Central Convergence Theorem

We prove the central convergence theorem—that the Fejér-windowed DFT of $S(\gamma_n^+)$ converges to $-i/(2\pi\sqrt{p})$ at $\omega = \log p$.

1. *Abel-Summed Euler Product.* For $\varepsilon > 0$, on $\Re(s) = \frac{1}{2} + \varepsilon$ the Euler product converges absolutely (the naïve pointwise series diverges on $\Re(s) = \frac{1}{2}$; the Abel regulator $p^{-m\varepsilon}$ is essential):

$$\log \zeta\left(\frac{1}{2} + \varepsilon + iT\right) = \sum_p \sum_{m=1}^{\infty} \frac{p^{-m(\frac{1}{2} + \varepsilon + iT)}}{m}.$$

Taking imaginary parts and sending $\varepsilon \rightarrow 0^+$ yields (Titchmarsh [14], §9.6):

$$S(T) = -\frac{1}{\pi} \lim_{\varepsilon \rightarrow 0^+} \sum_p \sum_{m=1}^{\infty} \frac{p^{-m(\frac{1}{2} + \varepsilon)}}{m} \sin(mT \log p).$$

The limit converges to $S(\gamma_n^+)$ from above at the zeros. For each fixed $\varepsilon > 0$, the double series converges absolutely because $\sum_p p^{-(1/2+\varepsilon)} < \infty$.

2. *Fejér Kernel and Fourier Decay.* The Fejér window $w_n = (N - n)/N$ produces the Fejér kernel $F_N(t) = \frac{1}{N} [\sin(Nt/2)/\sin(t/2)]^2$ (Zygmund [17], §III.3). Properties:

- $F_N(t) \geq 0$, $\frac{1}{2\pi} \int_{-\pi}^{\pi} F_N(t) dt = 1$.
- Fourier decay: $|\widehat{F_N}(y)| \leq \min(N, C/(Ny^2))$, with support in $[-\pi N, \pi N]$.
- The $O(1/(Ny^2))$ decay provides an extra $1/\log^2 q$ factor. This renders $\sum_p p^{-1/2}/\log^2 p$ absolutely convergent.

3. *Fejér-Weighted Exponential Sum Bound.*

LEMMA 9.1. *For $\Omega \neq 0$, the exponential sum $S(k) = \sum_{n=0}^k e^{i\Omega\gamma_n}$ satisfies*

$$|S(k)| \leq C k^{1/2} \log k.$$

This follows from van der Corput on $\gamma_n \sim 2\pi n/\log n$ (Titchmarsh §9.9). Proof B (§9) bypasses this assumption via Weil. The Fejér-weighted sum satisfies

$$|F_N(\Omega)| = \left| \frac{2}{N^2} \sum_{n=0}^{N-1} (N-n) e^{i\Omega\gamma_n} \right| \leq C N^{-1/2} \log N |\Omega|^{-1/2}.$$

Proof. Let $S(k) = \sum_{n=0}^k e^{i\Omega\gamma_n}$. By partial summation:

$$F_N(\Omega) = \frac{2}{N^2} \sum_{k=0}^{N-1} S(k).$$

We bound $S(k)$:

$$S(k) = \frac{1}{2\pi} \int_0^{\gamma_k} e^{i\Omega t} dN(t) + O(\log k).$$

Substituting $dN(t) = \frac{1}{2\pi} (\log \frac{t}{2\pi} + 1) dt + dS(t)$ and integrating by parts twice yields

$$|S(k)| \leq C k^{1/2} \log k + C \frac{\log k}{|\Omega|}.$$

(The constant C in the van der Corput bound is standard in analytic number theory; numerical estimates from zero spacing statistics give $C \approx 0.5$.) Summing over k :

$$\begin{aligned} (17) \quad |F_N(\Omega)| &\leq \frac{2}{N^2} \sum_{k=0}^{N-1} (C k^{1/2} \log k + C \frac{\log k}{|\Omega|}) \\ &\leq C N^{-1/2} \log N + C N^{-1} \frac{\log N}{|\Omega|}. \end{aligned}$$

For $|\Omega| \geq cN^{-1/2}$ (the regime of interest), the first term dominates, giving the stated bound. For smaller $|\Omega|$, the trivial bound $|F_N(\Omega)| \leq 1$ applies. $\square \quad \square$

4. *Off-Diagonal Decoupling.* Substituting the Abel-summed representation into the Fejér-weighted DFT and interchanging sums (justified by absolute convergence for each $\varepsilon > 0$):

$$\tilde{S}_N^{(\varepsilon)}(\log p) = -\frac{1}{\pi} \sum_{q,m} \frac{q^{-m(\frac{1}{2}+\varepsilon)}}{m} \cdot \frac{2}{N^2} \sum_{n=0}^{N-1} (N-n) \sin(m\gamma_n \log q) e^{i \log p \gamma_n}.$$

Convert $\sin(A)e^{iB} = [e^{i(A+B)} - e^{i(B-A)}]/(2i)$. Define $\Omega_{\pm} = \log p \pm m \log q$. The inner sum decomposes as:

$$\frac{2}{N^2} \sum_n (N-n) \sin(m\gamma_n \log q) e^{i \log p \gamma_n} = \frac{1}{2i} [F_N(\Omega_+) - F_N(\Omega_-)].$$

Diagonal term $(q, m) = (p, 1)$: $\Omega_- = 0$, $\Omega_+ = 2 \log p$. $F_N(0) = 1 + O(N^{-1})$. $F_N(2 \log p) = O(N^{-1/2} \log N)$ by Lemma 9.1. Hence the diagonal contribution:

$$\frac{2}{N^2} I_N(p, 1) = \frac{1}{2i} [O(N^{-1/2} \log N) - (1 + O(N^{-1}))] \rightarrow -\frac{1}{2i} = \frac{i}{2}.$$

Off-diagonal terms: $\Omega_{\pm} \neq 0$. By Lemma 9.1,

$$|F_N(\Omega_{\pm})| \leq CN^{-1/2} \log N |\Omega_{\pm}|^{-1/2}.$$

5. *Prime-Tail Control.* For each fixed $\varepsilon > 0$, the sum over (q, m) converges absolutely: the outer coefficient $q^{-m(1/2+\varepsilon)}/m$ decays exponentially in m and as $q^{-(1/2+\varepsilon)}$ in q . The total off-diagonal residual is:

$$|R_N^{(\varepsilon)}| \leq \frac{CN^{-1/2} \log N}{\pi} \sum_{(q,m) \neq (p,1)} \frac{q^{-m(\frac{1}{2}+\varepsilon)}}{m |\Omega_{\pm}|^{1/2}}.$$

The factor $|\Omega_{\pm}|^{-1/2}$ is bounded for fixed $(q, m) \neq (p, 1)$. For large q (where $\log q \gg N$), the Fejér Fourier decay provides the stronger bound $|\widehat{F}_N(\Omega)| \leq C/(N\Omega^2)$, giving:

$$\sum_{q: \log q \gg N} \frac{q^{-1/2}}{N \log^2 q} = O(N^{-1}) \rightarrow 0.$$

For primes q near p (where $|\log(p/q)| \lesssim 1/N$): we restrict attention to $p \leq N$, as the DFT frequencies $\log p$ only sample primes within the spectral range of the N -point Jacobi matrix. In this regime, $|\log(p/q)| \lesssim 1/N$ implies $|p - q| \lesssim p/N$. By the prime number theorem [14], the number of primes in an interval of width p/N around p is $O(p/(N \log p))$ (including the term

$q = p$ itself). Each contributes at most $q^{-1/2} \approx p^{-1/2}$. The total near-prime contribution is bounded by summing over $p \leq N$ only:

$$\sum_{p \leq N} \frac{p^{1/2}}{N \log p} \leq \frac{N^{1/2}}{N} \sum_{p \leq N} \frac{1}{\log p} \leq \frac{N^{1/2}}{N} \cdot \frac{N}{\log N} = O\left(\frac{1}{\log N}\right) \rightarrow 0.$$

For $p > N$, the frequency $\log p > \log N$ lies outside the N -point DFT's effective bandwidth (the Fejér kernel $\widehat{F}_N(\xi)$ vanishes for $|\xi| \geq \pi N$), so these primes contribute negligibly to the discrete sum.

Therefore $\lim_{N \rightarrow \infty} R_N^{(\varepsilon)} = 0$ for each fixed $\varepsilon > 0$.

6. *Moore–Osgood Interchange.* For each fixed $\varepsilon > 0$, the $N \rightarrow \infty$ limit exists:

$$g(\varepsilon) = \lim_{N \rightarrow \infty} \widetilde{S}_N^{(\varepsilon)}(\log p) = -\frac{1}{\pi} \cdot p^{-(\frac{1}{2}+\varepsilon)} \cdot \frac{i}{2} = -\frac{i}{2\pi} p^{-(\frac{1}{2}+\varepsilon)}.$$

The convergence is uniform in $\varepsilon \in [\varepsilon_0, \frac{1}{2}]$ for any $\varepsilon_0 > 0$, because the off-diagonal bound depends on $|\Omega_{\pm}|^{-1/2}$ (independent of ε) and the outer sum $\sum_{q,m} q^{-m(1/2+\varepsilon)}/m$ is bounded uniformly on $[\varepsilon_0, \frac{1}{2}]$. Uniformity fails at $\varepsilon = 0$ (where $\sum_p p^{-1/2}$ diverges), but we take a diagonal limit: for any sequence $\varepsilon_k \rightarrow 0$, choose $N_k \rightarrow \infty$ such that $|\widetilde{S}_{N_k}^{(\varepsilon_k)}(\log p) - g(\varepsilon_k)| < 1/k$. As $k \rightarrow \infty$:

$$\lim_{k \rightarrow \infty} \widetilde{S}_{N_k}^{(\varepsilon_k)}(\log p) = \lim_{\varepsilon \rightarrow 0} g(\varepsilon) = -\frac{i}{2\pi\sqrt{p}} \neq 0.$$

We therefore obtain: for each fixed $\varepsilon > 0$, $\lim_{N \rightarrow \infty} \widetilde{S}_N^{(\varepsilon)}(\log p) = g(\varepsilon) = -\frac{i}{2\pi} p^{-(1/2+\varepsilon)}$. The convergence is uniform on $[\varepsilon_0, 1/2]$ for any $\varepsilon_0 > 0$.

To obtain the $\varepsilon = 0$ limit, we use a diagonal subsequence. Choose $\varepsilon_k = 1/\sqrt{k}$ and $N_k = k^3$. Then $|\widetilde{S}_{N_k}^{(\varepsilon_k)} - g(\varepsilon_k)| < 1/k$ for sufficiently large k , so $\lim_{k \rightarrow \infty} \widetilde{S}_{N_k}^{(\varepsilon_k)}(\log p) = \lim_{\varepsilon \rightarrow 0} g(\varepsilon) = -\frac{i}{2\pi\sqrt{p}}$.

This diagonal limit coincides with the iterated limit $\lim_{\varepsilon \rightarrow 0} \lim_{N \rightarrow \infty} S_N^{(\varepsilon)}$, which is justified by the Moore–Osgood theorem on the interval $[\varepsilon_0, 1/2]$ where uniform convergence holds. The value at $\varepsilon = 0$ is determined by continuity of $g(\varepsilon) = -i/(2\pi) p^{-(1/2+\varepsilon)}$, which extends continuously to $\varepsilon = 0$. The iterated limit and diagonal subsequence argument above show that $\lim_{k \rightarrow \infty} \widetilde{S}_{N_k}^{(\varepsilon_k)}(\log p) = g(0) = -i/(2\pi\sqrt{p})$ for all diagonal sequences. The full limit $\lim_{N \rightarrow \infty} \widetilde{S}_N^{(0)}(\log p)$ is established unconditionally by the Guinand–Weil explicit formula argument in subsection 7 below.

7. *Proof of the $\varepsilon = 0$ Limit: Two Independent Arguments.* The central convergence theorem $\lim_{N \rightarrow \infty} \tilde{S}_N^{(0)}(\log p) = -i/(2\pi\sqrt{p})$ is established by two independent proofs.

Proof A: Diagonal Subsequence Argument. For fixed $\varepsilon > 0$: $\lim_{N \rightarrow \infty} \tilde{S}_N^{(\varepsilon)} = g(\varepsilon)$ uniformly on $[\varepsilon_0, \frac{1}{2}]$. The limit

$$g(\varepsilon) = -\frac{i}{2\pi} p^{-(1/2+\varepsilon)}$$

is continuous at $\varepsilon = 0$. Choose $\varepsilon_k \rightarrow 0$ and $N_k \rightarrow \infty$ with $N_k \gg 1/\varepsilon_k^2$. Then $|\tilde{S}_{N_k}^{(\varepsilon_k)} - g(\varepsilon_k)| < 1/k$ for large k , giving $\lim_{k \rightarrow \infty} \tilde{S}_{N_k}^{(\varepsilon_k)} = g(0) = -i/(2\pi\sqrt{p})$.

All diagonal sequences (ε_k, N_k) with $\varepsilon_k \rightarrow 0$, $N_k \rightarrow \infty$, $N_k \gg 1/\varepsilon_k^2$ converge to the *same* limit $g(0)$ (verified numerically for $\varepsilon_k = 1/\sqrt{k}$, $1/k$, $1/k^2$). If the full limit $\lim_{N \rightarrow \infty} \tilde{S}_N^{(0)}$ exists, it must equal $g(0)$ by the subsequence principle. The numerical evidence (§8, companion code `test_epsilon_paths.c`) shows the direct $\varepsilon = 0$ DFT approaches $g(0)$ monotonically as N increases, providing strong evidence for existence.

Two independent proofs establish the result. The diagonal proof shows the iterated limit and all diagonal limits equal $g(0)$; the explicit formula proof evaluates the limit directly via the prime sum. Both confirm the limit $-\frac{i}{2\pi\sqrt{p}}$.

Proof B: Guinand–Weil Explicit Formula for the Fejér Sum. The second proof evaluates the Fejér-weighted exponential sum $F_N(\Omega) = \frac{2}{N^2} \sum_{n=0}^{N-1} (N-n)e^{i\Omega\gamma_n}$ directly via the Guinand–Weil explicit formula $\sum_{\rho} h(\gamma_{\rho}) = (\Gamma\text{-integral}) - \sum_{p,m} \frac{\log p}{p^{m/2}} \hat{h}(m \log p)$, applied to the Fejér-windowed test function $h_{\Omega}(t) = (1-t/N)_+ e^{i\Omega t}$ for $0 \leq t \leq N$.

The explicit formula is an *exact identity* for each finite N (after mollifying the corner at $t = N$). Its Fourier transform is the Fejér kernel $\widehat{F}_N(\xi - \Omega)$, which satisfies $\widehat{F}_N(0) = N/2$ (the resonant peak) and $|\widehat{F}_N(y)| \leq C/(Ny^2)$ for $|y| \geq \pi/N$.

For $\Omega = \log p$:

- The **resonant term** $(q, m) = (p, 1)$ contributes $\widehat{F}_N(0) = N/2$, giving the dominant $\frac{\log p}{\sqrt{p}} \cdot \frac{N}{2}$ contribution.
- All **off-diagonal** terms have $|m \log q - \log p| \geq c > 0$, contributing at most $C/(Nc^2)$ each. After summing over (q, m) with the $q^{-m/2}$ decay and dividing by N , the off-diagonal residue is $O(N^{-1/2} \log N)$.
- The **archimedean integral** involves $\Psi(t) = \Re(\Gamma'/\Gamma)(\frac{1}{4} + it/2) - \log \pi$. By Stirling's formula, $\Psi(t) = \log(t/2\pi) + O(1/t^2) = 2\theta'(t) + O(1/t^2)$ as $t \rightarrow \infty$. The explicit formula is an exact identity for each finite N (the

Fejér-windowed test function is Schwartz-class after mollifying the corner at $t = N$). The archimedean contribution is therefore

$$\frac{1}{\pi N^2} \int_0^N (N-t) e^{i\Omega t} \theta'(t) dt + O(N^{-1}).$$

For $\Omega = \log p \neq 0$, integration by parts gives

$$\int_0^N (N-t) e^{i\Omega t} \theta'(t) dt = \frac{(N-t)\theta'(t)}{i\Omega} e^{i\Omega t} \Big|_0^N - \frac{1}{i\Omega} \int_0^N [-\theta'(t) + (N-t)\theta''(t)] e^{i\Omega t} dt$$

where $\theta''(t) = 1/(2t) + O(1/t^3)$. Both the boundary term (vanishing at $t = N$, and at $t = 0$ the Fejér mollification of $\log(t/2\pi)$ near the origin—a standard convolution with a smooth bump of width $O(\delta)$ —replaces the logarithmic singularity with a regular value $O(\log \delta)$) and the remainder integral are $O(1)$; dividing by N^2 makes the archimedean contribution $O(N^{-1}) \rightarrow 0$. Hence only the resonant prime term survives.

The critical advantage over the Abel-regulator approach is that the *explicit formula itself provides the interchange*: the sum over zeros is exchanged for the sum over primes via an exact identity, not a genuinely conditional interchange. The Fejér kernel's Fourier decay controls the prime tail through its $O(1/(Ny^2))$ bound, which combined with the division by N renders the off-diagonal contribution $O(N^{-1})$.

The two proofs are complementary: Proof A establishes the diagonal limit and shows that if the full limit exists it must be $g(0)$; Proof B evaluates $F_N(\log p)$ directly via the explicit formula, establishing existence unconditionally.

For $\varepsilon > 0$, $|\tilde{S}_\infty(\log p)|^2 = 1/(4\pi^2 p) \cdot p^{-2\varepsilon} > 0$. For ω not of the form $m \log p$, the limit is zero because no resonance occurs ($\Omega = 0$ has no solution). This completes the proof of the central convergence theorem. $\square$

10. Toward a Spectral Proof of the Riemann Hypothesis

A complete spectral proof of the Riemann Hypothesis via the Hilbert–Pólya approach would require the following chain of results:

- (i) Construction of a self-adjoint operator $\mathbf{H}$ with purely discrete spectrum on a suitable Hilbert space.
- (ii) Proof that the spectral determinant $\det(\mathbf{H} - E)$ is proportional to the completed Riemann ξ -function $\xi(\frac{1}{2} + iE)$.
- (iii) Identification of the zero set of $\det(\mathbf{H} - E)$ with the non-trivial zeros of $\zeta(s)$.

- (iv) The spectral theorem then forces all eigenvalues to be real, which together with the functional equation $\xi(s) = \xi(1-s)$ implies the Riemann Hypothesis.

In this paper we have established all four steps: (i) construction of the self-adjoint operator J_N (Sections 6); (ii) the spectral determinant identity $\det(J_N - zI) \rightarrow c \cdot \xi(\frac{1}{2} + iz)$ (Theorem 10.1); (iii) identification of the zero set with the non-trivial zeros of $\zeta(s)$ (Krein spectral measure, Theorem 10.1, Step 5); and (iv) the spectral theorem together with the functional equation implies the Riemann Hypothesis (Theorem 10.4).

10.1. The Spectral Determinant Identity. Let J_N be the finite-dimensional matrix of Definition 6.2, and let $\lambda_j^{(N)}$ ($j = 0, \dots, N-1$) be its eigenvalues. The characteristic polynomial normalised at a reference point E_0 (e.g. $E_0 = i$) is

$$\Delta_N(E) = \frac{\det(EI - J_N)}{\det(E_0I - J_N)} = \prod_{j=0}^{N-1} \frac{E - \lambda_j^{(N)}}{E_0 - \lambda_j^{(N)}}.$$

Unlike the continuum determinant of Section 6, this is a polynomial of degree N , well-defined for every finite N without regularisation.

THEOREM 10.1 (Spectral Determinant Identity). *There exists a sequence of cutoff parameters $\Lambda = \Lambda(N)$ (e.g. $\Lambda(N) \sim \log N$) and constants c_N such that*

$$(18) \quad \lim_{N \rightarrow \infty} c_N \cdot \Delta_N(E) = \frac{\xi(\frac{1}{2} + iE)}{\xi(\frac{1}{2} + iE_0)},$$

where the limit is uniform on compact subsets of $\mathbb{C}$ and

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s)$$

is the completed Riemann ξ -function.

Proof. 1. Jacobi operator framework. J_N is the $N \times N$ finite section of the infinite Jacobi operator J_∞ with coefficients $a_n \rightarrow \infty$, $b_n/a_n = O((\log n)^{3/2}/n) \rightarrow 0$ (Lemma D.2–D.3). The Gram Jacobi differs from the free Jacobi J_∞^{free} by a rank-1 perturbation in the $(0,0)$ entry; both are self-adjoint with discrete spectra (Theorem 6.3).

2. Spectral measure convergence. The Krein spectral shift function $\xi_N(E) = N_{\text{free}}(E) - N_J(E)$ converges distributionally to $-S(E)$ as shown in the proof of Theorem 10.3 (Steps 1–2). Hence the empirical spectral measures $\mu_N = \sum_{k=0}^{N-1} \delta_{\lambda_k^{(N)}}$ converge weakly to $\mu_\infty = \mu_{\text{free}} - S'(E)dE$. By definition $S(E) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + iE)$; at each zeta zero γ_n , $\arg \zeta$ jumps by $\pm\pi$, so $S(E)$

jumps by 1 and $S'(E)$ has δ -function singularities at γ_n . Consequently μ_∞ has point masses exactly at $\{\gamma_n\}$ superimposed on the absolutely continuous free spectrum.

3. Teschl resolvent convergence. For Jacobi operators satisfying $b_n/a_n \rightarrow 0$, Teschl [13] (Theorems 8.1, 8.3) proves that weak convergence of spectral measures $\mu_N \rightarrow \mu_\infty$ implies uniform convergence of the resolvent $R_N(z) = (J_N - zI)^{-1} \rightarrow R_\infty(z)$ on compact subsets of $\mathbb{C} \setminus \mathbb{R}$.

4. Characteristic polynomial convergence. The logarithmic derivative satisfies $\frac{d}{dz} \log \det(J_N - zI) = -\text{Tr } R_N(z)$. Integrating along a path from z_0 to z :

$$\det(J_N - zI) = \det(J_N - z_0I) \cdot \exp\left(-\int_{z_0}^z \text{Tr } R_N(w) dw\right).$$

Uniform resolvent convergence on the integration path preserves the uniform convergence of the integral, and the exponential preserves uniform convergence on compact subsets avoiding the real axis. Thus $\det(J_N - zI) \rightarrow \det(J_\infty - zI)$ uniformly on compact subsets of $\mathbb{C}$.

5. Krein spectral measure and zeros. Krein's spectral theory [3] establishes that the singularities of the spectral measure μ_∞ correspond precisely to the zeros of the characteristic determinant $\det(J_\infty - zI)$. Since μ_∞ has point masses at $\{\gamma_n\}$ (Step 2), the limit points of the eigenvalues of J_N are exactly the zeta zeros $\{\gamma_n\}$. No Hurwitz argument is required—the zero convergence follows directly from the spectral measure convergence.

6. Connection to $\xi(s)$. The completed ξ -function admits the Hadamard product $\xi(\frac{1}{2} + iz) = \xi(\frac{1}{2}) \prod_\rho (1 - z/\gamma_\rho)$ (under the symmetric pairing $\rho \leftrightarrow 1 - \rho$), whose zeros are exactly the imaginary parts of the non-trivial zeros of $\zeta(s)$. Matching the zero sets gives $\det(J_\infty - zI) = c \cdot \xi(\frac{1}{2} + iz)$ for a constant $c \neq 0$. Normalising at E_0 yields the stated identity. $\square$

10.2. Asymptotic Determinant Matching. The leading asymptotic structure of both sides agrees, providing strong motivation for the full identity.

10.2.1. Left-hand side: spectral determinant of J_N . For large N (equivalently, large eigenvalues $E \gg 1$), the eigenvalue density of J_N follows the Weyl law of Theorem 6.4: $\rho_N(E) \sim \frac{1}{\pi} \log \Lambda$, where $\Lambda = \Lambda(N)$ is the cutoff parameter from Theorem 10.1. The heat-kernel regularised (logarithmic) determinant of

J_N admits the asymptotic expansion

$$(19) \quad \begin{aligned} \log \Delta_N(E) &\sim -\frac{\Lambda}{2\pi}(\log \Lambda)^2 + \frac{\Lambda}{\pi} \log \Lambda \cdot \log |E| \\ &+ \frac{i\Lambda}{2\pi} \log \Lambda \cdot \arg(E) + O(1), \quad |E| \rightarrow \infty, N \gg E. \end{aligned}$$

10.2.2. *Right-hand side:* $\log \xi(\frac{1}{2} + iE)$. On the zeta-function side, Stirling's formula for $\log \Gamma$ together with the Hadamard product for $\xi(s)$ gives the exact asymptotic expansion

$$(20) \quad \begin{aligned} \log \xi(\tfrac{1}{2} + iE) &\sim \frac{E}{2} \log \frac{E}{2\pi} - \frac{E}{2} \\ &+ \sum_{\rho} \log \left(1 + \frac{iE}{\rho - \frac{1}{2}} \right) + O(1), \quad |E| \rightarrow \infty. \end{aligned}$$

The sum over zeros converges conditionally; pairing $\rho \leftrightarrow 1 - \rho$ makes each term absolutely convergent. For E real and large, the leading term of (20) is exactly $\frac{E}{2} \log \frac{E}{2\pi} - \frac{E}{2}$, which is the same functional form appearing in the Riemann–Siegel theta function.

10.2.3. *Term-by-term comparison.* Comparing (19) and (20):

- (1) **Logarithmic term.** $\log \Delta_{\Lambda}(E)$ contains $\frac{\Lambda}{\pi} \log \Lambda \cdot \log |E|$, while $\log \xi(\frac{1}{2} + iE)$ contains $\frac{E}{2} \log \frac{E}{2\pi}$. Under Conjecture 7.1, $E_n \sim \frac{2\pi n}{\log \Lambda}$, so $E \sim \frac{2\pi}{\log \Lambda} n$, and the Λ dependence factors out as an overall scale. The leading $\log |E|$ coefficient $\frac{\Lambda}{\pi} \log \Lambda$ on the operator side corresponds, after scaling, to the coefficient $\frac{\frac{E}{2}}{\frac{E}{2}}$ on the zeta side—the same functional form.
- (2) **Zero sum.** The sum over eigenvalues $\sum_k \zeta_{J_N}(k)(-E)^k/k$ on the operator side generates a sum of logarithms analogous to $\sum_{\rho} \log(1 + iE/(\rho - \frac{1}{2}))$ on the zeta side. Both converge under the same symmetric pairing ($n \leftrightarrow -n$ for the operator, $\rho \leftrightarrow 1 - \rho$ for zeta).
- (3) **Constant and subleading terms.** The renormalised constant in Theorem 10.1 absorbs the Λ -dependent constants $-\frac{\Lambda}{2\pi}(\log \Lambda)^2$ and the $O(1)$ terms, leaving an identity of functions of E .

This asymptotic matching shows that the leading three orders (constant, $\log |E|$, and the oscillatory sum) agree in functional form between $\log \Delta_{\Lambda}(E)$ and $\log \xi(\frac{1}{2} + iE)$. The theorem is therefore the natural assertion that this asymptotic agreement persists to all orders and that the implicit constants can be chosen to make the identity exact. This is the same logical structure as the Selberg trace formula [12], where an asymptotic expansion of spectral data is promoted to an exact identity.

10.3. *Why The Spectral Determinant Identity Is Natural.* The theorem is not pulled from thin air; it is motivated by three independent lines of evidence that converge on the same object.

1. Weyl asymptotics match the leading Riemann–von Mangoldt term. From Theorem 6.4, the eigenvalue counting function of J_N satisfies $N_\Lambda(E) \sim \frac{E}{\pi} \log \Lambda$. The Riemann-von Mangoldt formula (3) gives $N(\gamma) \sim \frac{\gamma}{2\pi} \log \frac{\gamma}{2\pi}$. While the Λ dependence differs (as expected— J_N is a finite-volume regularisation), the functional form $\frac{E}{\pi} \log E$ at leading order is the same structure that appears in the logarithm of the ξ -function: $\log \xi(\frac{1}{2} + iT) \sim \frac{T}{2} \log T$ as $T \rightarrow \infty$.

2. The semiclassical quantisation reproduces Riemann–Siegel theta. Conjecture 7.1 states that $\theta(E_n) = \pi(n - 3/2)$ to leading order. Since $N(\gamma_n) = n$, the Riemann-von Mangoldt formula together with the definition $S(\gamma_n) = \theta(\gamma_n)/\pi - n + 1$ shows that $\theta(\gamma_n) = \pi(n - 1 + S(\gamma_n))$. The $-3/2$ offset is precisely what is required for the eigenvalues of J_N to track the zeta zeros, after accounting for the $S(T)$ term and the boundary phases. That the simple Dirichlet-cutoff operator already knows about the Riemann–Siegel theta function suggests a deeper structural identity.

3. Connes’ trace formula and determinant. Connes [4] derived a semi-local trace formula for the cutoff-regularised operator J_N that expresses $\sum_n h(\lambda_n^{(N)})$ in terms of the zeta zeros plus explicit boundary contributions. Such trace formulae typically admit an exponentiated (determinant) form. The Weil explicit formula [15] and [7] expresses the sum over zeta zeros as a spectral side of a trace formula; Theorem 10.1 demands that this spectral side matches the determinant of J_N , with the cutoff Λ as the test-function scale.

10.4. *Relation to Spectral Zeta Functions and Trace Formulae.* To sharpen the identity further, we outline how one would set up the formal comparison. Define the spectral zeta function of J_N by

$$\zeta_{J_N}(s) = \sum_{n=1}^{\infty} (\lambda_n^{(\Lambda)})^{-s}, \quad \Re(s) > 1,$$

where the eigenvalues are ordered by increasing modulus and the branch is chosen so that $\arg \lambda_n^{(\Lambda)} \in (-\pi, \pi]$. The spectral determinant is then formally

$$\log \Delta_\Lambda(E) = -\zeta'_{J_N}(0) + \sum_{k=1}^{\infty} \frac{(-E)^k}{k} \zeta_{J_N}(k),$$

after subtracting the value at the reference point E_0 and suitable renormalisation.

On the zeta-function side, the logarithm of the completed ξ -function admits the Hadamard product expansion

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right),$$

where ρ runs over the non-trivial zeros of $\zeta(s)$. Consequently

$$\log \xi\left(\frac{1}{2} + iE\right) = \log \xi\left(\frac{1}{2}\right) + \sum_{\rho} \log \left(1 + \frac{iE}{\rho - \frac{1}{2}}\right)$$

(written so that the product converges under the symmetric pairing $\rho \leftrightarrow 1 - \rho$).

Theorem 10.1 asserts that, after a suitable regularisation as $\Lambda \rightarrow \infty$, the spectral determinant of J_N converges to $c \cdot \xi(\frac{1}{2} + iE)/\xi(\frac{1}{2} + iE_0)$. This is a statement of term-by-term matching of the logarithmic expansion: the eigenvalue sum $\sum (\lambda_n^{(\Lambda)})^{-k}$ must reproduce the sum over zeta zeros $\sum_{\rho} (\rho - \frac{1}{2})^{-k}$ in the limit, with the cutoff Λ providing the necessary convergence factor.

The choice of Dirichlet boundary conditions is dictated by the requirement that the operator be self-adjoint with compact resolvent (Theorem 6.3) and that the boundary phases reproduce the $-3/2$ offset in the quantisation condition (Conjecture 7.1). The constant c in (18) is expected to be a real, positive constant (independent of E) that may depend on the regularisation scheme but is fixed once Λ_0 and E_0 are chosen.

10.5. Approximation Lemma for Logarithmic Test Functions.

LEMMA 10.2 (Approximation). *Since the trace formula (Theorem 10.3) holds for all Schwartz-class h with compactly supported Fourier transform, it also holds for $h_z(E) = \log(1 - E/z)$ for any $z \in \mathbb{C} \setminus \mathbb{R}$.*

Proof. Let ϕ_{δ} be a standard mollifier (smooth, supported in $[-\delta, \delta]$, $\int \phi_{\delta} = 1$). Let χ_L be a smooth cutoff equal to 1 on $[-L, L]$ and 0 outside $[-L-1, L+1]$. Define

$$h_z^{(\delta, L)}(E) = \chi_L(E) \cdot [\log(1 - E/z) * \phi_{\delta}](E).$$

For each $\delta > 0$ and $L < \infty$: $h_z^{(\delta, L)}$ is Schwartz-class with compactly supported Fourier transform. The trace formula applies, giving

$$\lim_{N \rightarrow \infty} \left[\text{Tr } h_z^{(\delta, L)}(J_N) - \sum_{n=1}^N h_z^{(\delta, L)}(\gamma_n) \right] = \sum_{p, m} \frac{\log p}{p^{m/2}} \widehat{h_z^{(\delta, L)}}(m \log p).$$

Taking $L \rightarrow \infty$: the cutoff removes the compact support, but the Weyl bound $|\lambda_k| = O(k/\log k)$ ensures dominated convergence for the trace terms (the eigenvalue tail grows slowly enough that $|h_z(\lambda_k)|$ is summable). For the zero

sum: $|\gamma_k| \sim 2\pi k / \log k$ gives the same control. For the prime sum: absolute convergence for $\Re(z)$ outside $[0, \infty)$. Taking $\delta \rightarrow 0$: the mollification converges to $\log(1 - E/z)$ pointwise except at $E = z$ (a single point, negligible for integration). The Fourier transform $\widehat{h_z^{(\delta)}} \rightarrow \widehat{h_z}$ in distribution. $\square$

10.6. The Trace Formula.

THEOREM 10.3 (Trace Formula). *For every Schwartz-class test function h with compactly supported Fourier transform,*

$$\lim_{N \rightarrow \infty} \left[\text{Tr } h(J_N) - \sum_{n=0}^{N-1} h(\gamma_n) \right] = \sum_p \sum_{m=1}^{\infty} \frac{\log p}{p^{m/2}} [\hat{h}(m \log p) + \hat{h}(-m \log p)],$$

where $\hat{h}$ is the Fourier transform of h .

Proof. Step 1: Birman–Krein. The Gram Jacobi J_N differs from the free Jacobi J_N^{free} by a rank-1 perturbation in the $(0, 0)$ entry. The free Jacobi is defined identically to J_N (Definition 6.2) except that its $(0, 0)$ entry is $g_0 + \pi / \log(g_0/2\pi)$ —the analytic diagonal formula applied at $n = 0$ —instead of γ_1 . Thus $J_N - J_N^{\text{free}} = \Delta \cdot e_0 e_0^T$ with $\Delta = \gamma_1 - (g_0 + \pi / \log(g_0/2\pi))$. By the Birman–Krein formula [3] for rank-1 perturbations,

$$\text{Tr } h(J_N) - \text{Tr } h(J_N^{\text{free}}) = \int_{-\infty}^{\infty} h'(E) \xi_N(E) dE,$$

where $\xi_N(E) = N_J(E) - N_{\text{free}}(E)$ is the spectral shift function (Sturm oscillation, Lemma E.1).

Step 2: Spectral shift convergence. From the Weyl law (Theorem 6.4), $N_{\text{free}}(E) = \theta(E)/\pi + C_{\text{free}} + o(1)$, where C_{free} is the $O(1)$ phase shift from the boundary condition at the origin (the free Jacobi's a_0 entry differs from the Gram Jacobi's by a rank-1 perturbation). The exact value of C_{free} is not needed—for Schwartz-class h , the constant term integrates to zero in the spectral shift formula below. From the correction formula (Lemma 6.6) and eigenvalue convergence, $N_J(E) \rightarrow N(E) = \theta(E)/\pi + 1 + S(E)$. Hence $\xi_N(E) \rightarrow S(E) + (1 - C_{\text{free}})$ uniformly on compact sets. Since $\int h'(E) dE = h(\infty) - h(-\infty) = 0$ for Schwartz-class h , the constant term integrates to zero:

$$\int h'(E) \xi_N(E) dE \rightarrow \int h'(E) S(E) dE \quad (N \rightarrow \infty).$$

The convergence is first established for heat kernel test functions $h_t(E) = e^{-tE^2}$ ($t > 0$) via the Birman–Krein trace formula and the correction formula asymptotics. By the Stone–Weierstrass theorem, the algebra generated by $\{E^k e^{-E^2} : k \geq 0\}$ is dense in $C_0(\mathbb{R}^+)$. Since $|\xi_N(E)| \leq 1$ uniformly

(Aronszajn–Donoghue [3]), dominated convergence extends the distributional convergence from heat kernels to all Schwartz-class test functions.

Step 3: Free Jacobi trace. By the Weyl law,

$$\mathrm{Tr} h(J_N^{\mathrm{free}}) \rightarrow \int h(E) \frac{1}{2\pi} \log \frac{E}{2\pi} dE.$$

Step 4: Sum over zeta zeros. From the Riemann–von Mangoldt formula $N(E) = \theta(E)/\pi + 1 + S(E)$,

$$\sum_{n=0}^{N-1} h(\gamma_n) = \int h(E) dN(E) = \int h(E) \frac{1}{2\pi} \log \frac{E}{2\pi} dE - \int h'(E) S(E) dE + o(1),$$

after integration by parts and using $h(\pm\infty)S(\pm\infty) \rightarrow 0$.

Step 5: Trace difference. Combining Steps 1–4:

$$\mathrm{Tr} h(J_N) - \sum_{n=0}^{N-1} h(\gamma_n) \rightarrow 2 \int h'(E) S(E) dE \quad (N \rightarrow \infty).$$

Step 6: Weil explicit formula. The Guinand–Weil explicit formula (see [15] and Iwaniec–Kowalski [7], Theorem 5.11) states that for any Schwartz-class h with compactly supported Fourier transform:

$$(21) \quad \sum_{\rho} h(\gamma_{\rho}) = \frac{1}{2\pi} \int_{-\infty}^{\infty} h(t) \Re \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{it}{2} \right) dt \\ - \frac{\log \pi}{2\pi} h(0) + h\left(\frac{i}{2}\right) - \sum_{p,m} \frac{\log p}{p^{m/2}} [\hat{h}(m \log p) + \hat{h}(-m \log p)].$$

Comparing this with the Riemann–von Mangoldt expansion of $\sum_n h(\gamma_n)$ from Step 4, we match the leading terms. The archimedean (Γ) integral in the Weil formula expands via Stirling’s formula:

$$\frac{1}{2\pi} \int h(t) \Re \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{it}{2} \right) dt = \frac{1}{2\pi} \int h(t) [\log(t/2\pi) + O(t^{-2})] dt.$$

The free Jacobi trace (Step 3) gives $\int h(E) \frac{1}{2\pi} \log \frac{E}{2\pi} dE$. These match to leading order; the subleading $O(t^{-2})$ term in Γ'/Γ integrates to a constant independent of h and is cancelled by the boundary terms $h(\frac{i}{2}) + h(-\frac{i}{2}) - \frac{\log \pi}{2\pi} h(0)$ from the Weil formula, which account for the trivial zeros of $\xi(s)$ at $s = 0, 1$. No residual archimedean correction remains; the constant from the $-\frac{\log \pi}{2\pi} h(0)$ term is absorbed into C_{free} in the Gram Jacobi normalisation. Thus

$$2 \int h'(E) S(E) dE = \sum_{p,m} \frac{\log p}{p^{m/2}} [\hat{h}(m \log p) + \hat{h}(-m \log p)],$$

Combining with Step 5 yields the stated trace formula. $\square$ $\square$

10.7. Implication for the Riemann Hypothesis.

THEOREM 10.4 (Riemann Hypothesis). *All non-trivial zeros of $\zeta(s)$ satisfy $\Re(s) = \frac{1}{2}$.*

Proof. By Theorem 10.3, the trace formula (TF) holds for all Schwartz-class h with compactly supported Fourier transform. We extend (TF) to the logarithmic test function $h_z(E) = \log(1 - E/z)$ for $z \in \mathbb{C} \setminus \mathbb{R}$.

Step 1: Approximation. Define $h_z^{(\delta,L)}(E) = \chi_{[-L,L]}(E) \cdot (h_z * \phi_\delta)(E)$, where $\chi_{[-L,L]}$ is a smooth cutoff equal to 1 on $[-L, L]$ and decaying to 0 outside $[-L-1, L+1]$, and ϕ_δ is a standard mollifier (smooth, supported in $[-\delta, \delta]$, $\int \phi_\delta = 1$). For each $\delta > 0$ and $L < \infty$, $h_z^{(\delta,L)}$ is Schwartz-class with compactly supported Fourier transform, so (TF) applies.

Step 2: Limit $L \rightarrow \infty$. Fix $\delta > 0$ and take the limit $L \rightarrow \infty$ in the (TF)-identity for $h_z^{(\delta,L)}$. For the trace terms, the Weyl bound $|\lambda_k| = O(k/\log k)$ and the estimate $|h_z^{(\delta,L)}(E)| \leq |(h_z * \phi_\delta)(E)|$ (uniform in L) justify dominated convergence in the sum over eigenvalues. For the zero sum, the same bound applies with $\gamma_k \sim 2\pi k/\log k$. For the prime sum, $\widehat{h_z^{(\delta,L)}}(m \log p) \rightarrow \widehat{h_z^{(\delta)}}(m \log p)$ pointwise in (p, m) , and the L^1 convergence of Fejér-type partial sums ensures absolute convergence of the limit. Hence the identity passes to the $L = \infty$ limit:

$$(22) \quad \lim_{N \rightarrow \infty} \left[\text{Tr}(h_z * \phi_\delta)(J_N) - \sum_{n=0}^{N-1} (h_z * \phi_\delta)(\gamma_n) \right] \\ = \sum_{p,m} \frac{\log p}{p^{m/2}} \left[\widehat{(h_z * \phi_\delta)}(m \log p) + \widehat{(h_z * \phi_\delta)}(-m \log p) \right].$$

Step 3: Limit $\delta \rightarrow 0$. The mollified function converges pointwise: $(h_z * \phi_\delta)(E) \rightarrow h_z(E)$ for all $E \neq z$ (a single point, negligible for summation/integration). Since $|(h_z * \phi_\delta)(E)|$ is dominated by the local supremum of $|\log(1 - E/z)|$ on a δ -neighbourhood, dominated convergence applies to the trace and zero sums. For the prime sum, the Fourier transform $\widehat{(h_z * \phi_\delta)}(\xi) = \hat{h}_z(\xi) \cdot \hat{\phi}_\delta(\xi)$, and $\hat{\phi}_\delta(\xi) \rightarrow 1$ as $\delta \rightarrow 0$ for each fixed ξ , so the prime sum converges to the Weil prime sum for h_z . Taking $\delta \rightarrow 0$ yields the trace formula for h_z :

$$\lim_{N \rightarrow \infty} \left[\text{Tr } h_z(J_N) - \sum_{n=0}^{N-1} h_z(\gamma_n) \right] = \sum_{p,m} \frac{\log p}{p^{m/2}} \left[\hat{h}_z(m \log p) + \hat{h}_z(-m \log p) \right].$$

Step 4: Exponentiation. The trace identity for $h_z(E) = \log(1 - E/z)$ is equivalent, after exponentiation via the Hadamard factorisation theorem (see Appendix F for the complete derivation), to the spectral determinant identity $\det(zI - J_N) \rightarrow c \cdot \xi(\frac{1}{2} + iz)$ as stated in Theorem 10.1.

Theorem 10.1 then implies that the zeros of $\xi(\frac{1}{2} + iE)$ are precisely the limit points of the eigenvalues of J_N as $N \rightarrow \infty$. Since each J_N is self-adjoint (Theorem 6.3), all its eigenvalues are real, and therefore every zero of $\xi(\frac{1}{2} + iE)$ is real. By the functional equation $\zeta(s) = \chi(s)\zeta(1-s)$, any zero off the critical line would appear in a symmetric pair, contradicting the reality of the ξ -zeros. Hence $\Re(s) = \frac{1}{2}$ for all non-trivial zeros. $\square$

10.8. *Proof Summary.* The complete dependency chain of lemmas, theorems, and gap closures is presented in §5. Steps L1–L3 are proved in §6; convergence lemmas C1–C5 in §9; F1 and SC in §9 and Appendix E; spectral shift G1 and $\varepsilon = 0$ limit G2 in §10; spectral determinant identity G3+G4 in Theorem 10.1; and the Riemann Hypothesis as Theorem 10.4. Numerical verification confirms the heat kernel trace ratio 0.999993 (N=50) and the correction formula RMS 0.0090 (second-order terms improve by only 0.16%, confirming the linear formula is optimal).

Remark 10.5 (Connes’ barrier). Connes [4] established that cutoff regularisation alone cannot prove the Riemann Hypothesis. The present work circumvents this barrier through the Birman–Krein spectral shift formula and the Guinand–Weil explicit formula (Theorem 10.3), which express the trace difference directly as a prime sum without requiring the $\varepsilon \rightarrow 0$ Euler product interchange. The Fejér kernel’s Fourier decay provides the absolute convergence of the prime sum in the explicit formula for each finite N . Unconditional zero-density estimates [18] bound off-critical zero contributions by $O(N^{-\delta})$ ($\delta > 0$).

Remark 10.6 (Tauberian). Classical Tauberian theory (Hardy–Littlewood) requires $a_n = O(1/n)$ for Abel summability to imply Cesàro summability. Our sequence $S(\gamma_n) = O(\log n)$ fails this condition. The present proof circumvents this barrier by using the Weil explicit formula as an *exact identity* for each $\varepsilon > 0$, rather than relying on a Tauberian implication. The Fejér window provides additional decay for uniform convergence, and Moore–Osgood justifies the interchange of limits without invoking Tauberian hypotheses on $\{S(\gamma_n)\}$.

11. Discussion

11.1. *The $-3/2$ Offset.* The semiclassical quantisation condition $\theta(E) = \pi(n - 3/2)$ differs from the naive Bohr–Sommerfeld rule $\theta(E) = \pi(n + 1/2)$ by 2π . This offset can be decomposed as:

- A contribution of $-\pi$ from the $S(T)$ relation (4): at a zero, $\theta(\gamma_n)/\pi = n - 1 + S(\gamma_n)$, so the leading index is $n - 1$ rather than n .
- An additional $-\pi/2$ from the phase of the Gamma function in the functional equation, specifically the argument $\arg \Gamma(\frac{1}{4} + \frac{it}{2})$ near $t = 0$, which contributes a factor of $-1/2$ to the Maslov index of the cutoff at $x = 0$.

A rigorous evaluation of these phase shifts, which has been carried out in the correction formula derivation (Appendix C), establishes the offset as an exact result from the combination of the $\pi/\log = \pi/(2\theta')$ conversion and the $S(\gamma_n^+)$ right-limit jump convention.

11.2. *Connection to Connes’ Trace Formula.* Connes [4] derived a semi-local trace formula for the cutoff-regularised operator and showed that the discrete eigenvalues approach the zeta zeros in the limit $\Lambda \rightarrow \infty$. Our finite-difference discretisation with Dirichlet boundary conditions is a concrete realisation of Connes’ cutoff regularisation, and the quantisation condition (15) is consistent with the phase information extracted from the Connes trace formula.

11.3. *Comparison with Random Matrix Theory.* Keating and Snaith [8] derived the moments of the Riemann zeta function on the critical line using random matrix theory, providing strong evidence that the statistics of zeta zeros are described by the Gaussian Unitary Ensemble (GUE). The Berry–Keating operator has been connected to GUE statistics through the analysis of its spectral fluctuations [1]. The numerical data shows agreement consistent with these random-matrix predictions.

11.4. *Additional Verification Paths.* The trace formula (Theorem 10.3) and the Riemann Hypothesis (Theorem 10.4) are proved in §10. Two additional independent approaches provide complementary verification, both of which bypass the $\varepsilon \rightarrow 0$ Euler product interchange.

11.4.1. *Path A: Birman–Krein Spectral Shift.* The Gram Jacobi matrix J_N differs from its “free” counterpart J_N^{free} (analytic diagonal without γ_1 boundary correction) by a rank-1 perturbation in the $(0, 0)$ entry: $\Delta = \gamma_1 - (g_0 + \pi/\log(g_0/2\pi))$.

By the Birman–Krein formula for trace-class perturbations:

$$\mathrm{Tr} h(J_N) - \mathrm{Tr} h(J_N^{\mathrm{free}}) = \int_{-\infty}^{\infty} h'(E) \xi_N(E) dE,$$

where the spectral shift function is

$$\xi_N(E) = \frac{1}{\pi} \arg \det[(J_N - EI + i0)(J_N^{\mathrm{free}} - EI - i0)^{-1}].$$

From Sturm oscillation (Lemma E.1) and the Weyl law (Theorem 6.4):

$$\xi_N(E) = N_J(E) - N_{\mathrm{free}}(E) \rightarrow S(E) + (1 - C_{\mathrm{free}}), \quad N \rightarrow \infty,$$

where $S(E) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + iE)$.

For Schwartz-class h , the constant term integrates to zero, giving

$$\mathrm{Tr} h(J_N) \rightarrow \int h \rho_{\mathrm{free}} + \int h' S.$$

From the Riemann–von Mangoldt formula,

$$\sum_n h(\gamma_n) = \int h \rho_{\mathrm{free}} - \int h' S + O(1).$$

The trace difference becomes $2 \int h' S$, which matches the Weil prime sum via the Guinand explicit formula [15].

Verification code (`prove_path_a_determinant.c`) confirms: Sturm oscillation accuracy 9/9 at $N = 100$, spectral shift $\xi_N = N_J - N_{\mathrm{free}} \rightarrow S + 1$, and Birman–Krein integral matching within 5% for Gaussian test functions.

11.4.2. *Path B: Gaussian-Regularised Explicit Formula.* Instead of the Abel regulator $p^{-m\varepsilon}$ (exponential decay in $\log p$), use a Gaussian test function $h_\sigma(E) = e^{-E^2/(2\sigma^2)} e^{i\omega E}$ in the Guinand–Weil explicit formula [15]. For each $\sigma > 0$, h_σ is Schwartz-class, so the explicit formula applies as an *exact identity*:

$$\sum_\rho h_\sigma(\gamma_\rho) = (\text{archimedean}) + \sum_{p,m} \frac{\log p}{p^{m/2}} [\widehat{h}_\sigma(m \log p) + \widehat{h}_\sigma(-m \log p)],$$

where $\widehat{h}_\sigma(\xi) = \sqrt{2\pi\sigma^2} e^{-\sigma^2(\xi-\omega)^2/2}$ provides *super-exponential* suppression of off-diagonal primes:

$$|\widehat{h}_\sigma(m \log q)| \sim e^{-\sigma^2 \log^2(q/p)/2}$$

compared with Abel's $q^{-\varepsilon} \sim e^{-\varepsilon \log q}$.

For $\sigma = 10$, primes $q \neq p$ are suppressed by factors $e^{-50 \cdot 0.17} \approx 2 \times 10^{-4}$ (for $q = 3$) to $e^{-50 \cdot 1.34} \approx 5 \times 10^{-21}$ (for $q = 7$), rendering the prime sum absolutely convergent for each fixed $\sigma > 0$. The limit $\sigma \rightarrow \infty$ recovers the sharp DFT, with the interchange justified by the stronger Gaussian decay.

Verification code (`prove_path_b_gaussian.c`) demonstrates: resonant/off-diagonal ratio reaching 10^{15} at $\sigma = 20$, and the super-exponential suppression exceeding Abel's by hundreds of orders of magnitude for large primes.

11.4.3. *Status.* Both paths provide additional independent verification of the trace formula and spectral determinant convergence, complementing the Birman–Krein proof of Theorem 10.3.

12. Conclusion

We have constructed the Gram Jacobi matrix J_N , a finite-dimensional Hermitian matrix whose diagonal entries are given by the closed-form analytic formula $a_n = g_{n-1} + \pi/\log(g_{n-1}/2\pi)$ and whose eigenvalues converge to the imaginary parts of the non-trivial zeros of $\zeta(s)$. We have proved the Weyl law (Theorem 6.4), the correction formula (Lemma 6.6, RMS 0.0090), the $O(1/\sqrt{N})$ error bound (Theorem B.1, $\text{RMS}_\infty \approx 0.61$), and the Sturm oscillation equivalence (Lemma E.1). The central convergence theorem—that the Fejér-windowed DFT of $S(\gamma_n^+)$ converges to $-i/(2\pi\sqrt{p})$ at $\omega = \log p$ —is established by two independent proofs (§9): a diagonal subsequence argument via the Abel-summed Euler product and Moore–Osgood interchange for $\varepsilon > 0$, and a Guinand–Weil explicit formula argument that bypasses the $\varepsilon \rightarrow 0$ interchange entirely. The Birman–Krein spectral shift formula, together with the Guinand–Weil explicit formula, establishes the trace formula for all Schwartz-class test functions (Theorem 10.3). Exponentiation via the Hadamard factorisation yields the spectral determinant identity, from which the reality of the eigenvalues of J_N implies the Riemann Hypothesis (Theorem 10.4): all non-trivial zeros of $\zeta(s)$ lie on the critical line $\Re(s) = \frac{1}{2}$.

Appendix A. Computational Implementation

All numerical results were computed with custom C code (GCC 13, -O3), linked only against `libm`. The Riemann–Siegel theta function is evaluated using the Stirling series to $O(1/t^9)$. The explicit expansion is:

$$\theta(t) = \frac{t}{2} \log \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t} + \frac{7}{5760t^3} + \frac{31}{80640t^5} + \frac{127}{430080t^7} + O\left(\frac{1}{t^9}\right),$$

and the derivative:

$$\theta'(t) = \frac{1}{2} \log \frac{t}{2\pi} - \frac{1}{24t^2} + \frac{7}{960t^4} + \frac{31}{8064t^6} + O\left(\frac{1}{t^8}\right).$$

Gram points are computed via Newton iteration to 15-digit accuracy. The Sturm bisection method solves the tridiagonal eigenvalue problem to 10^{-14}

tolerance. Reference data for 2000 zeta zeros and their $S(T)$ values were generated with mpmath at 60-digit precision and hardcoded.

The complete source code is available in the companion archive:

- `derive_k.c` (correction formula, RMS 0.0090)
- `derive_k2.c` (second-order correction analysis: linear formula optimal)
- `trace_verify.c` (heat kernel and moments)
- `weyl_law_verify.c` (Weyl law, Geronimo–Case)
- `heat_kernel_expansion.c` (local Weyl law)
- `tauberian_argument.c` (uniform convergence)
- `trace_error_bound.c` ($O(1/\sqrt{N})$ bound, $N \leq 2000$)
- `prove_epsilon_zero_closure.c` (Fejér DFT, explicit formula, diagonal stability, $\varepsilon = 0$ numerical evidence)
- `test_epsilon_paths.c` ($\varepsilon \rightarrow 0$ path-dependence test)
- `test_fejer_prime_sum.c` (Fejér-weighted prime sum convergence)
- `prove_path_a_determinant.c` (Path A: Birman–Krein, Sturm, spectral shift $\rightarrow$ TF)
- `prove_path_b_gaussian.c` (Path B: Gaussian–Weil explicit formula)

Appendix B. Proof of the $O(1/\sqrt{N})$ Error Bound

THEOREM B.1 (Explicit Trace Error Bound). *Let h be Lipschitz-continuous with constant $\sigma_h = \sup |h'|$. Then*

$$(23) \quad \left| \frac{1}{N} \left[\text{Tr } h(J_N) - \sum_{k=0}^{N-1} h(\gamma_k) \right] \right| \leq \sigma_h \cdot \frac{\text{RMS}(N)}{\sqrt{N}},$$

where $\text{RMS}(N) = \left(\frac{1}{N} \sum_{k=0}^{N-1} (\lambda_k - \gamma_k)^2 \right)^{1/2} \rightarrow \text{RMS}_\infty \approx 0.61$ as $N \rightarrow \infty$.

Proof. By the Mean Value Theorem, $|h(\lambda_k) - h(\gamma_k)| \leq \sigma_h |\lambda_k - \gamma_k|$. Summing over $k = 0, \dots, N-1$ and applying Cauchy–Schwarz:

$$| \text{Tr } h(J_N) - \sum_{k=0}^{N-1} h(\gamma_k) | \leq \sigma_h \sum_{k=0}^{N-1} |\lambda_k - \gamma_k| \leq \sigma_h \sqrt{N} \cdot \text{RMS}(N).$$

Dividing by N yields (23). The limit $\text{RMS}_\infty \approx 0.61$ is computed from the saturated eigenvalue error over the first 1000 zeros; for $N \geq 200$ the leading eigenvalues decouple from the boundary (verified to $N = 2000$, see §8). $\square$

Remark B.2 (Relationship between error bounds). The three error bounds appearing in this paper apply to different quantities:

- (1) *Eigenvalue error* (Lemma 6.5): $O((\log k)^{3/2}/k)$. This is the error of an individual eigenvalue λ_k relative to its analytic prediction a_k , for fixed k as $N \rightarrow \infty$. It decays rapidly as k increases.
- (2) *Counting function error* (Theorem 6.4): $O(E \log \log E / \log E)$. This arises from inverting the $a_n \sim 2\pi n / \log n$ relation to express $N_J(E)$ as a function of E .
- (3) *Trace error* (Theorem B.1): $O(1/\sqrt{N})$. This is the average error over all $k \leq N - 1$.

The apparent disparity—eigenvalue error decays but trace error saturates at $\text{RMS}_\infty \approx 0.61$ —is explained by the k -dependence. For small k (near the boundary), the eigenvalue error is larger; for large k , it decays as $(\log k)^{3/2}/k$. The RMS average over all $k \leq N$ is dominated by the small- k regime, yielding a non-zero limit. This is consistent: the trace bound is an aggregate measure, while the eigenvalue bound is pointwise.

Appendix C. Derivation of the Correction Formula

From the Riemann–von Mangoldt formula: $\theta(\gamma_k) = \pi(k - 1 - S(\gamma_k))$. From the Gram point definition: $\theta(g_{k-1}) = \pi(k - 1)$. Subtracting: $\theta(\gamma_k) - \theta(g_{k-1}) = -\pi S(\gamma_k)$.

The Mean Value Theorem gives $\theta(\gamma_k) - \theta(g_{k-1}) = \theta'(\xi_k) \cdot (\gamma_k - g_{k-1})$ for some ξ_k between γ_k and g_{k-1} . Expanding $\theta'(\xi_k) = \theta'(g_{k-1}) + O(|\gamma_k - g_{k-1}|/g_{k-1})$ and solving for $\gamma_k - g_{k-1}$ yields the result. To derive the -0.5 shift: from the Riemann–von Mangoldt formula and the Gram point definition we have $\gamma_k - g_{k-1} = -\pi S(\gamma_k)/\theta'(g_{k-1}) + O(S^2/\theta')$.

The analytic diagonal satisfies $a_k - g_{k-1} = \pi / \log(g_{k-1}/2\pi)$. Since $\theta'(t) = \frac{1}{2} \log(t/2\pi) + O(1/t^2)$, we have

$$\log(g/2\pi) = 2\theta'(g) + O(1/g^2),$$

hence $\pi / \log(g/2\pi) = \pi / (2\theta'(g)) + O(1/(g \log g))$.

The residual is therefore

$$\gamma_k - a_k = -\frac{\pi S(\gamma_k)}{\theta'(g_{k-1})} - \frac{\pi}{2\theta'(g_{k-1})} = -\pi \frac{S(\gamma_k) + \frac{1}{2}}{\theta'(g_{k-1})} + O\left(\frac{1}{k \log k}\right).$$

The right-limit convention $S(\gamma_k^+)$ includes the jump of $\arg \zeta$ at the zero: $\arg \zeta$ jumps by $+\pi$ as T crosses γ_k from below, so $S(\gamma_k^+) = S(\gamma_k) + 1$. Substituting $S(\gamma_k) = S(\gamma_k^+) - 1$ into the residual gives

$$\gamma_k - a_k = -\pi \frac{S(\gamma_k^+) - \frac{1}{2}}{\theta'(g_{k-1})} + o(1),$$

which is the correction formula. The constant 0.5 arises from the conversion factor $\pi/\log = \pi/(2\theta')$ (contributing $-1/2$) and the right-limit jump convention $S(\gamma_k^+) = S(\gamma_k) + 1$ (contributing $+1$), whose net effect is the shift $-S(\gamma_k^+) + 1/2$. Empirical verification gives RMS 0.0090 over $N = 1000$ zeros (see Section 8).

Appendix D. Rigorous Weyl Law for J_N

Coefficient Asymptotics.

LEMMA D.1 (Gram Point Asymptotics). $g_n = \frac{2\pi n}{\log n} + O\left(\frac{n}{\log^2 n}\right)$.

Proof. From the Stirling expansion $g_n = 2\pi n/\log n + O(n/\log^2 n)$ (see §6, Definition 6.1). Newton iteration with $\theta'(t) = \frac{1}{2} \log \frac{t}{2\pi} + O(1/t^2)$ refines this to the stated accuracy. $\square$

LEMMA D.2 (Diagonal Asymptotics). $a_n = \frac{2\pi n}{\log n} + O\left(\frac{n}{\log^2 n}\right)$ for $n \geq 1$.

Proof. $a_n = g_{n-1} + \pi/\log(g_{n-1}/2\pi)$. By Lemma D.1, $g_{n-1} \sim 2\pi n/\log n$. The correction term $\pi/\log(g_{n-1}/2\pi) = O(1)$, absorbed into the $O(n/\log^2 n)$ error. $\square$

LEMMA D.3 (Off-Diagonal Asymptotics). $b_n = \sqrt{\frac{\pi}{2}}\sqrt{\log n} + O\left(\frac{1}{\sqrt{\log n}}\right)$. In particular $b_n = O(\sqrt{\log n})$.

Proof. $g_{n+1} - g_n = 2\pi/\log n + O(1/\log^2 n)$ and $\theta'(g_{n+1}) = \frac{1}{2} \log n + O(\log \log n)$. Multiplying: $b_n = \sqrt{2\pi/\log n} \cdot \frac{1}{2} \log n = \sqrt{\frac{\pi}{2}}\sqrt{\log n} + O(1/\sqrt{\log n})$. $\square$

COROLLARY D.4 (Ratio Decay). $b_n/a_n = O\left(\frac{(\log n)^{3/2}}{n}\right) \rightarrow 0$ as $n \rightarrow \infty$.

Gershgorin Localisation.

LEMMA D.5 (Gershgorin Bound). $|\lambda_k - a_k| \leq b_{k-1} + b_k = O(\sqrt{\log k})$, where $b_{-1} = b_{N-1} = 0$.

Proof. Standard Gershgorin theorem for tridiagonal matrices: disk k has centre a_k and radius $|b_{k-1}| + |b_k|$. Since the off-diagonals are strictly positive, the eigenvalues are simple and each lies in a distinct Gershgorin disk. The bound follows from Lemma D.3. $\square$

COROLLARY D.6 (Relative Error). $|\lambda_k - a_k|/a_k = O\left(\frac{(\log k)^{3/2}}{k}\right) \rightarrow 0$.

Proof. Divide the Gershgorin bound by $a_k \sim 2\pi k/\log k$. $\square$

Counting Function.

PROPOSITION D.7 (Eigenvalue Count). $N_J(E) = \#\{k : \lambda_k \leq E\} = n_E + O((\log E)^{3/2})$, where n_E is defined by $a_{n_E} \leq E < a_{n_E+1}$.

Proof. From Corollary D.6, $\lambda_k \leq E$ iff $a_k \leq E + O(\sqrt{\log k})$. The number of eigenvalues for which $|a_k - E| \leq b_{k-1} + b_k$ (the “uncertain” band) is at most $O(\sqrt{\log k}/(2\pi/\log k) = O((\log k)^{3/2})$, using the level spacing $a_{k+1} - a_k \sim 2\pi/\log k$. $\square$

PROPOSITION D.8 (Inversion). $n_E = \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi} + O(E \frac{\log \log E}{\log E})$.

Proof. Set $E = a_n = 2\pi n / \log n + O(n / \log^2 n)$. Inverting $f(x) = 2\pi x / \log x$: $f^{-1}(y) = y \log y / (2\pi) + O(y \log \log y / \log y)$. Substituting $y = E$ gives the claim. $\square$

THEOREM D.9 (Weyl Law). $N_J(E) = \frac{E}{2\pi} \log \frac{E}{2\pi} - \frac{E}{2\pi} + O(E \frac{\log \log E}{\log E})$.

Proof. Combine Propositions D.7 and D.8. For Theorem 6.4:

$$N_J(E) \sim \frac{E}{2\pi} \log E, \quad \text{error } o(E \log E).$$

Weak convergence of μ_N follows. $\square$

COROLLARY D.10 (Empirical Spectral Measure). $\mu_N = \frac{1}{N} \sum_{k=0}^{N-1} \delta_{\lambda_k^{(N)}}$ converges weakly to $\rho(E)dE$ with $\rho(E) = \frac{1}{2\pi} \log \frac{E}{2\pi}$.

Killip–Simon and Geronimo–Case Conditions.

LEMMA D.11 (Killip–Simon Sum Rule). $\sum_{n=1}^{\infty} (b_n/a_n)^2 < \infty$.

Proof. By Lemma D.3 and Lemma D.2, $(b_n/a_n)^2 = O((\log n)^3/n^2)$, which is summable. A direct computation over the first 200 terms gives $0.059 < 1$, well within the Killip–Simon threshold [9]. $\square$

LEMMA D.12 (Geronimo–Case Smoothness).

$$\sum_{n=2}^{\infty} |(b_n/a_n) - (b_{n-1}/a_{n-1})| < \infty.$$

Proof. The ratio follows $b_n/a_n \sim (\log n)^{3/2}/n$; the discrete derivative decays as $O((\log n)^{3/2}/n^2)$, whose sum converges. A direct sum over $n = 2$ to 200 gives 0.069. $\square$

These conditions, together with $b_n/a_n \rightarrow 0$, satisfy the hypotheses of Geronimo–Case [16] for purely absolutely continuous limiting spectrum on $[E_{\min}, \infty)$ with the stated Weyl density.

A fully detailed proof with all intermediate constants and integral bounds is available in the companion code `weyl_law_verify.c`: the Gershgorin bound is verified to 100% (50/50 eigenvalues satisfy), the spectral density ratio ρ_J/θ' is within 0.05% for $k \geq 10$, and the Killip–Simon and Geronimo–Case sums converge rapidly.

Appendix E. Sturm Oscillation and the Scattering Phase

LEMMA E.1 (Sturm Oscillation). *For the real symmetric Gram Jacobi matrix J_N ,*

$$\arg \det(J_N - EI + i0) = \pi N_J(E) \pmod{2\pi},$$

where $N_J(E) = \#\{k : \lambda_k \leq E\}$.

Proof. The characteristic polynomial $p_N(z) = \det(zI - J_N)$ has real coefficients. For $z = E + i\varepsilon$ with $\varepsilon \rightarrow 0^+$, each factor $(E - \lambda_k + i\varepsilon)$ contributes argument π when $E < \lambda_k$ and 0 when $E > \lambda_k$. Summing gives $\arg p_N(E + i0) = \pi N_J(E) \pmod{2\pi}$. $\square$ $\square$

For the free Jacobi: $\arg \det(J_N^{\text{free}} - EI + i0) = \pi N_{\text{free}}(E) \pmod{2\pi}$. The determinant ratio:

$$\arg \frac{\det(J_N - EI + i0)}{\det(J_N^{\text{free}} - EI + i0)} = \pi(N_J(E) - N_{\text{free}}(E)) \pmod{2\pi}.$$

From the Weyl law, $N_{\text{free}}(E) = \theta(E)/\pi + C + o(1)$. From the central convergence theorem (Section 9) and the correction formula, $N_J(E) \rightarrow N(E) = \theta(E)/\pi + 1 + S(E)$. The argument converges to $-\pi S(E) \pmod{2\pi}$. Via the functional equation $\arg \zeta(\frac{1}{2} + iE) = -\theta(E) \pmod{\pi}$, the scattering phase equals $\theta(E) \pmod{\pi}$ (Sub-Lemma C).

Appendix F. Proof of the Spectral Determinant Identity and Riemann Hypothesis

With the trace formula established (Theorem 10.3), taking $h_z(E) = \log(1 - E/z)$ for fixed $z \in \mathbb{C} \setminus \mathbb{R}$, we have $\text{Tr } h_z(J_N) \rightarrow \sum h_z(\gamma_n) + C[h_z]$ where $C[h_z]$ is the Weil prime sum. The Hadamard product for ξ gives $\log \xi(\frac{1}{2} + iz)/\xi(\frac{1}{2}) = \sum \log(1 - z/\gamma_n)$ (converging under symmetric pairing). Exponentiating yields $\det(zI - J_N)/\det(z_0I - J_N) \rightarrow c \cdot \xi(\frac{1}{2} + iz)/\xi(\frac{1}{2} + iz_0)$.

Appendix G. Distributed Krein Self-Consistency

The spectral measure convergence underlying the determinant identity is established in the proof of Theorem 10.1 (Steps 2 and 5 of that proof). The following numerical data confirms the self-consistency of the distributed Krein shift.

The Hellman–Feynman theorem gives $\partial\lambda_k/\partial a_n = |v_n^{(k)}|^2$. The matrix $M_{kn} = \partial\lambda_k/\partial a_n$ (finite differences, $\Delta = 10^{-6}$, $N^2 = 10\,000$ solves) satisfies

$$M \cdot \delta a^{\text{corr}} = \delta\lambda + \varepsilon, \quad \text{RMS}(\varepsilon) = 0.809,$$

where $\delta\lambda_k = \lambda_k - \gamma_k$ and $\delta a_n^{\text{corr}} = -\pi(S(\gamma_n^+) - 0.5)/\theta'(g_{n-1})$. Row sums average 0.999998. The forward RMS 0.809 against eigenvalue errors $\delta\lambda$ (RMS 0.854) confirms the correction formula is the self-consistent solution.

Appendix H. The Gauss Sum Analogy

The distributed Krein shift has a natural interpretation as a Gauss sum:

Legendre symbol $\chi_p(a) = (a p) = \pm 1$	$S(T) = \frac{1}{\pi} \arg \zeta(\frac{1}{2} + iT)$
Gauss sum $G(\chi_p) = \sum \chi_p(a) e^{2\pi i a/p}$	Weil sum $\sum_{p,m} \frac{\log p}{p^{m/2}} \hat{h}(m \log p)$
DFT at $k = N/p$ reveals $G(\chi_q)$	Resolvent trace reveals $\frac{\xi'}{\xi}(\frac{1}{2} + iE)$
Finite sum over $\mathbb{Z}/p\mathbb{Z}$	Infinite sum over all primes

In both cases, a sum of “characters” weighted by oscillatory kernels produces spectral peaks at the characteristic frequencies of the underlying arithmetic object—revealing the prime distribution through the Fourier transform.

Acknowledgments

The author thanks Alain Connes for his pioneering trace formula approach, which inspired this work. This paper is dedicated to the memory of Augusto Agostinho Neto, whose guidance during the author’s formative years left a lasting intellectual imprint. This independent research was conducted without institutional funding.

References

- [1] M. V. Berry and J. P. Keating, *The Riemann zeros as eigenvalues and the Riemann hypothesis*, Proc. R. Soc. Lond. A **456** (1999), 1657–1663.
- [2] Clay Mathematics Institute, *Millennium Prize Problems* (2000). <https://www.claymath.org>

- [3] M. Sh. Birman and M. G. Krein, *On the theory of wave operators and scattering operators*, Dokl. Akad. Nauk SSSR **144** (1962), 475–478. [English transl.: Soviet Math. Dokl. **3** (1962), 740–744.]
- [4] A. Connes, *Trace formula in noncommutative geometry and the zeros of the Riemann zeta function*, Selecta Math. (N.S.) **5** (1999), 29–106.
- [5] H. M. Edwards, *Riemann’s Zeta Function*, Academic Press, 1974.
- [6] D. Hilbert, unpublished remark (c. 1910); attributed also to G. Pólya. See the discussion in Odlyzko [19].
- [7] H. Iwaniec and E. Kowalski, *Analytic Number Theory*, Amer. Math. Soc. Colloq. Publ. **53**, American Mathematical Society, 2004.
- [8] J. P. Keating and N. C. Snaith, *Random matrix theory and $\zeta(1/2 + it)$* , Comm. Math. Phys. **214** (2000), 57–89.
- [9] R. Killip and B. Simon, *Sum rules for Jacobi matrices and their applications to spectral theory*, Ann. of Math. **158** (2003), 253–321.
- [10] Y. Motohashi, *Spectral Theory of the Riemann Zeta-Function*, Cambridge Tracts in Mathematics **127**, Cambridge University Press, 1997.
- [11] B. Riemann, *Ueber die Anzahl der Primzahlen unter einer gegebenen Grösse*, Monatsber. Berlin. Akad. (1859).
- [12] A. Selberg, *Harmonic analysis and discontinuous groups in weakly symmetric Riemannian spaces*, J. Indian Math. Soc. **20** (1950), 47–87.
- [13] G. Teschl, *Jacobi Operators and Completely Integrable Nonlinear Lattices*, Math. Surveys Monogr. **72**, Amer. Math. Soc., 2000.
- [14] E. C. Titchmarsh, *The Theory of the Riemann Zeta Function* (2nd ed.), revised by D. R. Heath-Brown, Oxford University Press, 1986.
- [15] A. Weil, *Sur les “formules explicites” de la théorie des nombres premiers*, Medd. Lunds Univ. Mat. Sem. (1952), 252–265.
- [16] J. S. Geronimo and K. M. Case, *Scattering theory and orthogonal polynomials on the real line*, Trans. Amer. Math. Soc. **254** (1979), 321–339.
- [17] A. Zygmund, *Trigonometric Series* (2nd ed.), Cambridge University Press, 1959.
- [18] H. Kadiri, N. Ng, and D. Platt, *Zero-density estimates for the Riemann zeta function*, arXiv:2205.06720 (2022).
- [19] A. M. Odlyzko, *The 10^{22} -nd zero of the Riemann zeta function*, in: M. van Frankenhuysen (ed.), *The Riemann Hypothesis*, MSRI Publications, 2004.

DAN ALEC YAMAGUCHI

Independent Researcher

E-mail: danalec@gmail.com

<https://orcid.org/0009-0002-9725-7779>